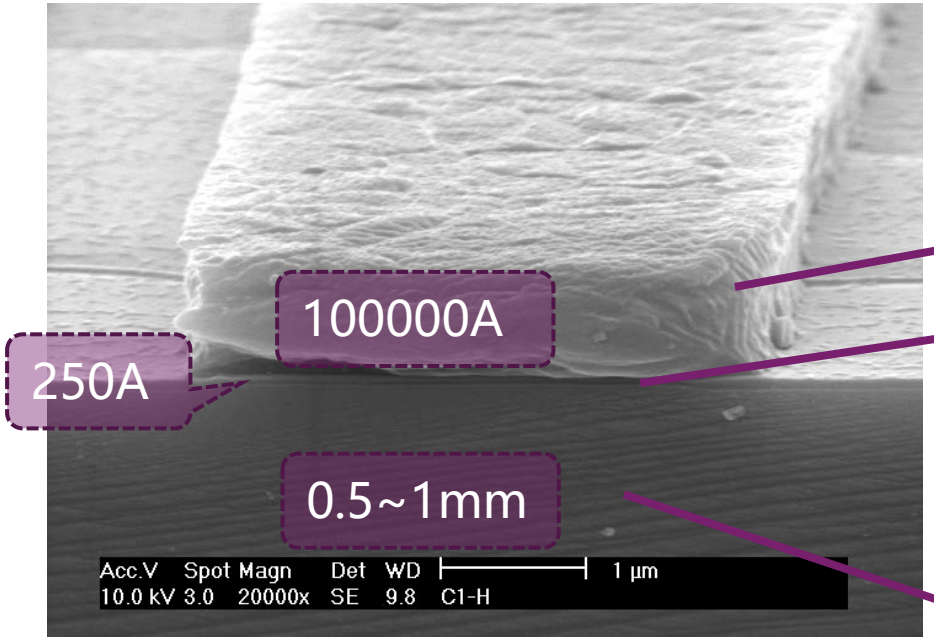
Chapter 10.
Fundamentals of MOSFET

PREVIEW:

1. Study the characteristics of energy bands as a function of applied voltage in the metal–oxide–semiconductor structure known as the MOS capacitor. The MOS capacitor is the heart of the MOSFET.
2. Discuss the concept of surface inversion in the semiconductor of the MOS capacitor.
3. Define and derive the expression for the threshold voltage, which is a basic pa rameter of the MOSFET.
4. Discuss various physical structures of MOSFETs, including enhancement and depletion mode devices.
5. Derive the ideal current–voltage relationship of the MOSFET.
6. Develop the small-signal equivalent circuit of the MOSFET. This circuit is used to relate small-signal currents and voltages in analog circuits.
7. Derive the frequency limiting factors of the MOSFET.

10.1THE TWO-TERMINAL MOS STRUCTURE



Metal: aluminum、polycrystalline silicon

Oxide: SiO_2
t_{ox}: the thickness of the oxide
ϵ_{ox}: the permittivity of the oxide

Semiconductor: silicon

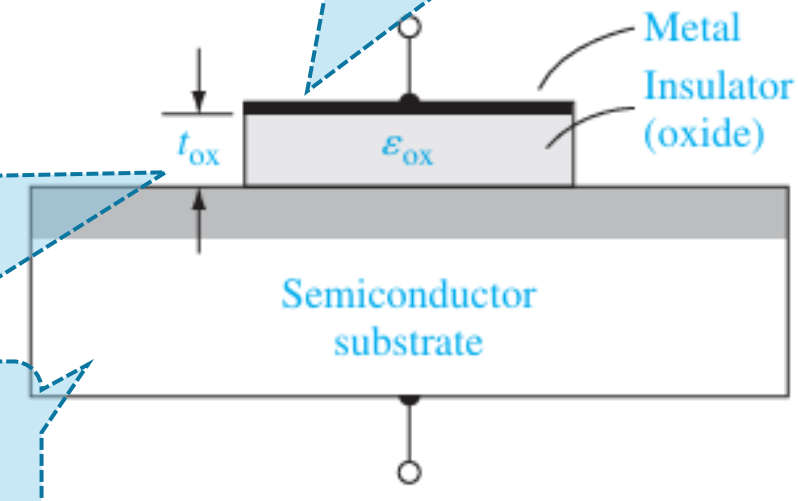


Figure 10.1 | The basic MOS capacitor structure.

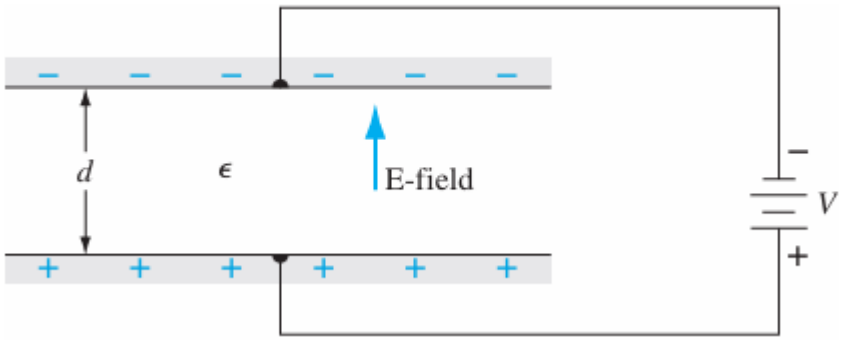
10.1.1 Energy-Band Diagrams

Parallel-plate Capacitor:

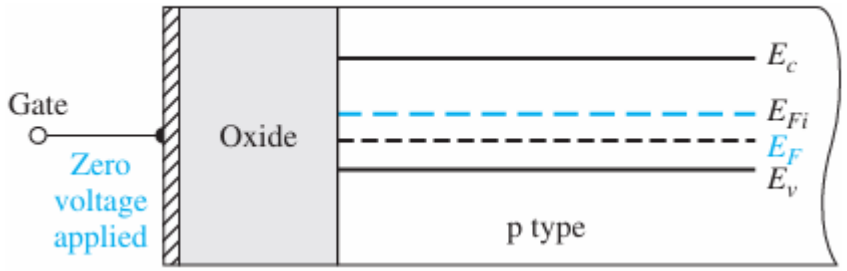
$$C' = \frac{\epsilon}{d}$$

$$Q' = C'V$$

$$E = \frac{V}{d}$$

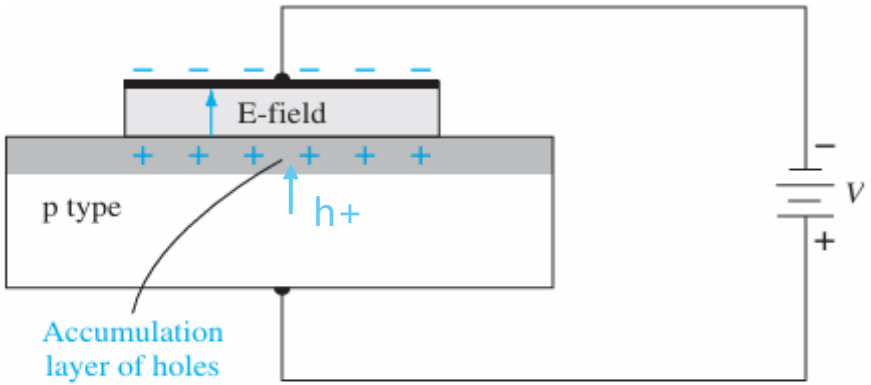


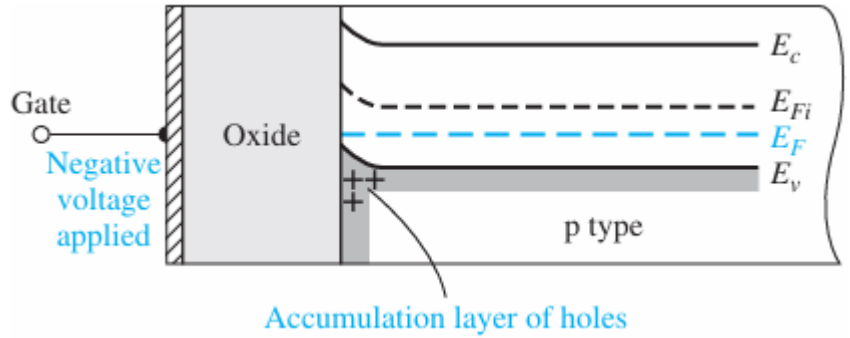
(1) MOS capacitor with p-type substrate with a zero gate bias



10.1.1 Energy-Band Diagrams

(2)MOS capacitor with p-type substrate with a negative gate bias

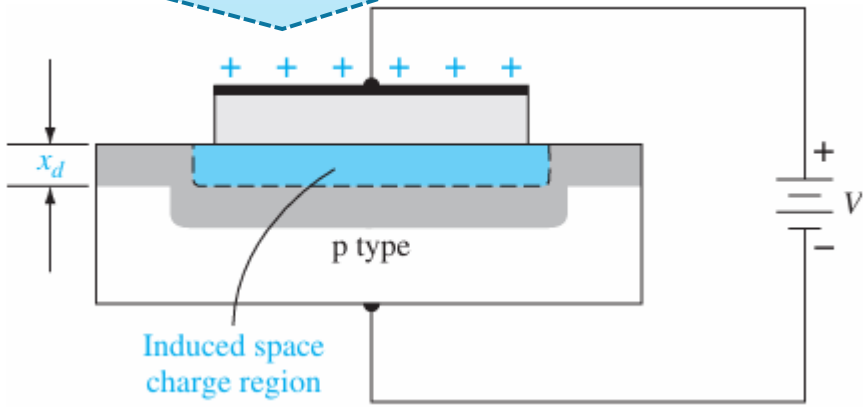


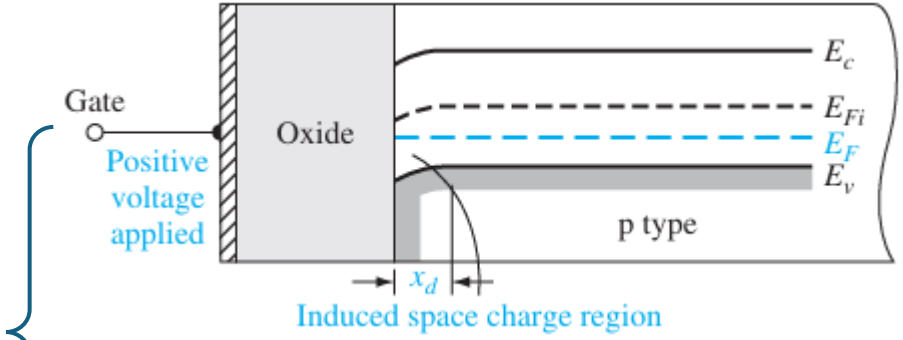


$(E_{FS} - E_v)\downarrow$

$E_{FS} \rightarrow E_v$

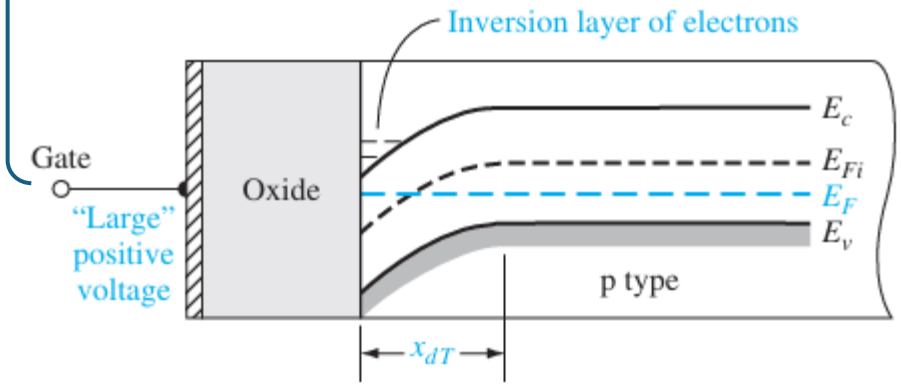
(3)MOS capacitor with p-type substrate with a positive gate bias





$(E_{FS} - E_v)\uparrow$

$E_{FS} \rightarrow E_{Fi}$

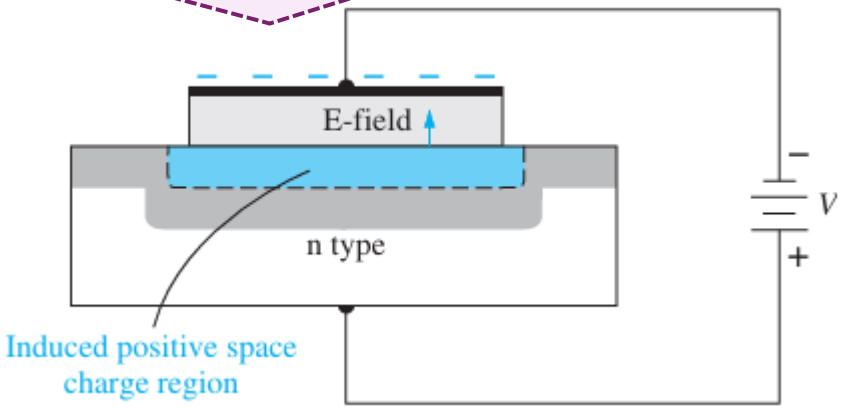


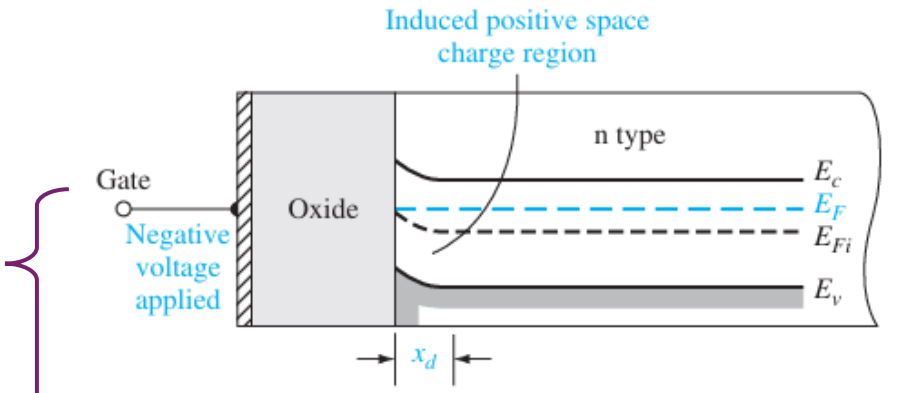
$(E_{FS} - E_v)\uparrow$

$E_{FS} > E_{Fi}$

10.1.1 Energy-Band Diagrams

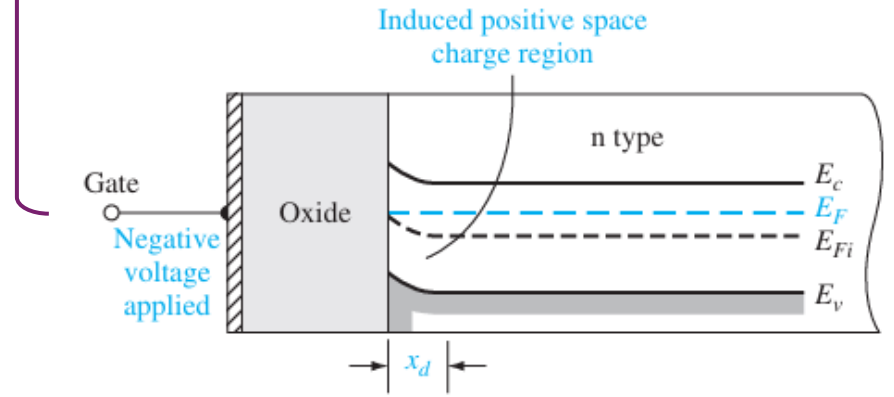
(4)MOS capacitor with n-type substrate with a negative gate bias





$(E_{FS} - E_c)\uparrow$

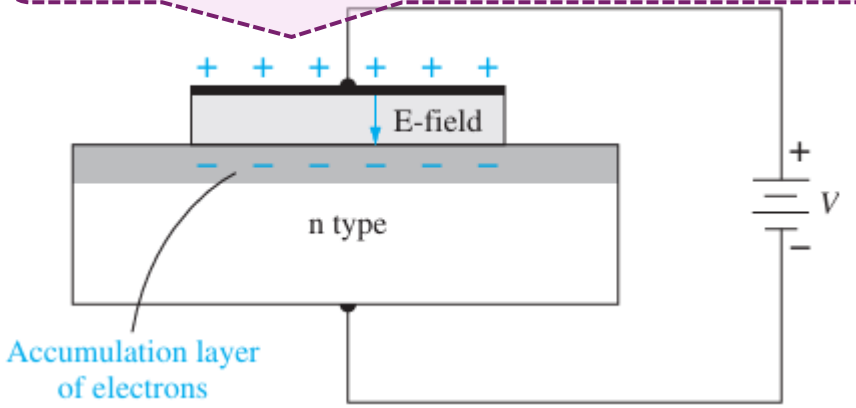
$E_{FS} \rightarrow E_{Fi}$

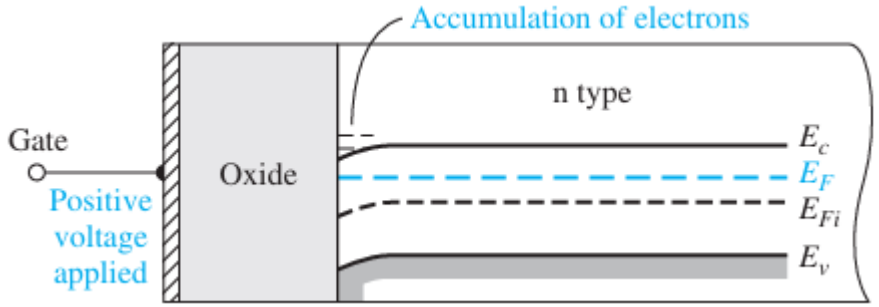


$(E_{FS} - E_c)\uparrow$

$E_{FS} < E_{Fi}$

(5)MOS capacitor with n-type substrate with a positive gate bias





$(E_{FS} - E_c)\downarrow$

$E_{FS} \rightarrow E_C$

10.1.2Depletion Layer Thickness

1.Fermi potential ϕ_{fp}

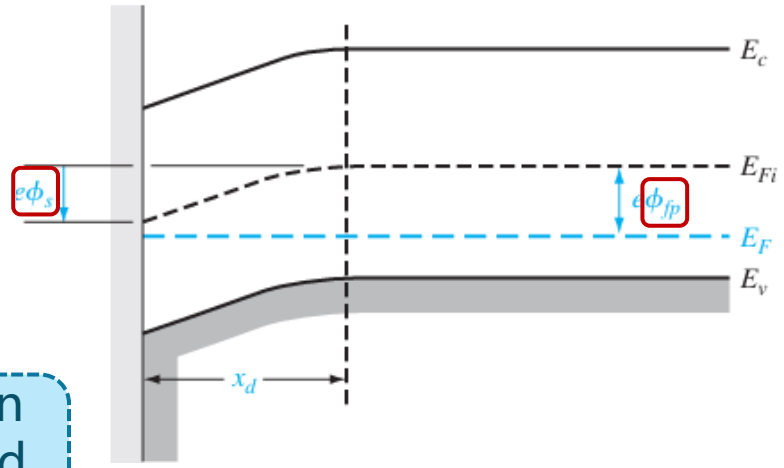
Difference (in V) between E_{Fi} and E_F

$$\phi_{fp} = V_t ln\left(\frac{N_a}{n_i}\right)$$

$$\phi_{fn} = V_t ln\left(\frac{N_d}{n_i}\right)$$

2.Surface potential ϕ_s

Difference (in V) between E_{Fi} measured in the bulk semiconductor and E_{Fi} measured at the surface

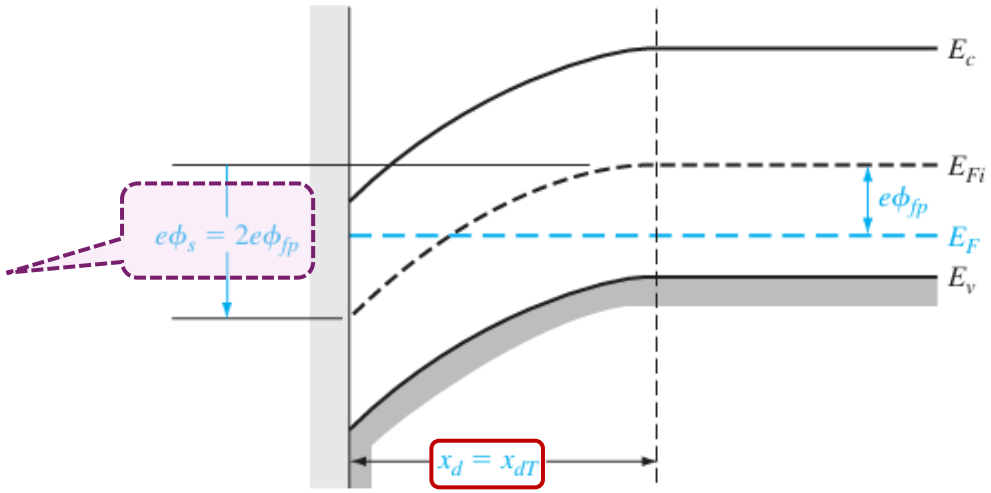


3.Depletion layer thickness

$$x_d = \left(\frac{2\epsilon_s\phi_s}{eN_a}\right)^{1/2}$$

$$\begin{cases} x_{dT} = \left(\frac{4\epsilon_s\phi_{fp}}{eN_a}\right)^{1/2} \\ x_{dT} = \left(\frac{4\epsilon_s\phi_{fn}}{eN_a}\right)^{1/2} \end{cases}$$

Maximum space charge width

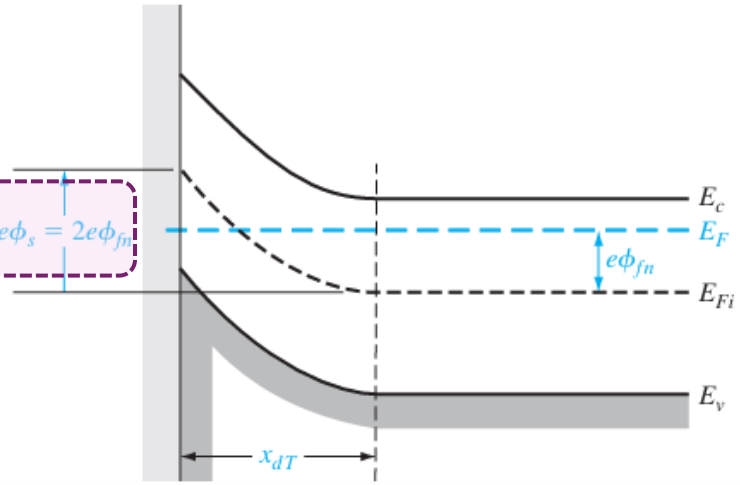


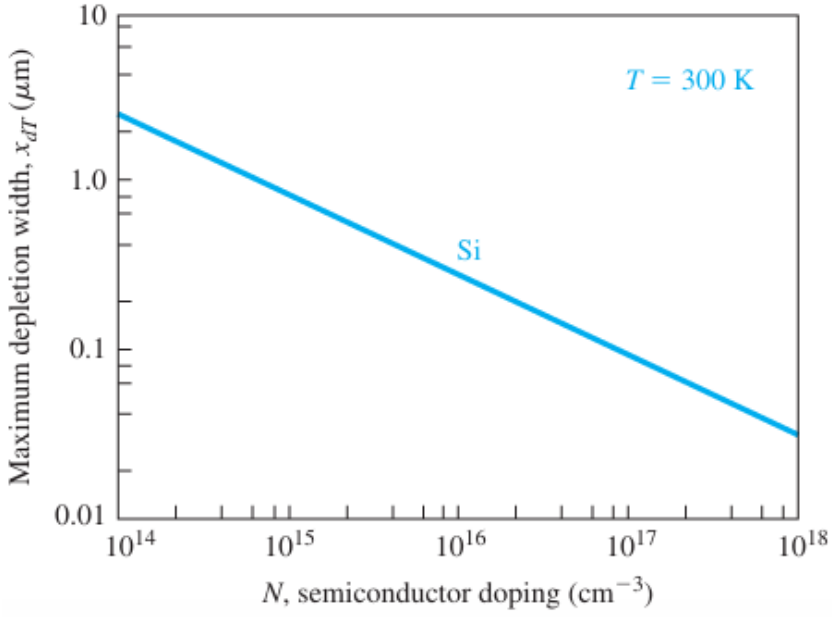
10.1.2Depletion Layer Thickness

$$x_d = \left(\frac{2\epsilon_s \phi_s}{eN_d}\right)^{1/2}$$

$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fn}}{eN_d}\right)^{1/2}$$

Maximum space charge width





10.1.3 Surface Charge Density

P-type substrate

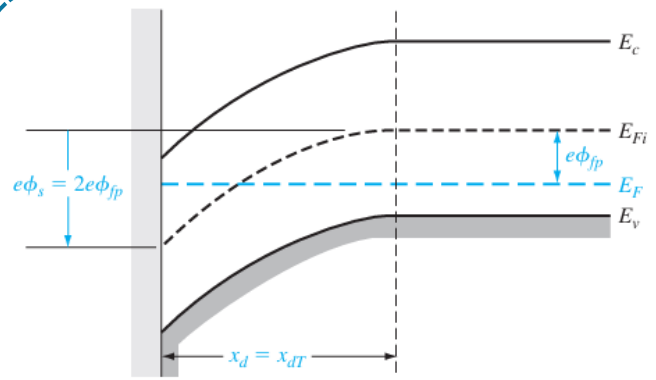
$$n_s = n_i exp\left[\frac{e(\phi_{fp}+\Delta\phi_s)}{kT}\right] = n_i exp\left[\frac{\phi_{fp}+\Delta\phi_s}{V_t}\right]$$

$$n = n_i exp\left(\frac{E_F - E_{Fi}}{kT}\right)$$

$$n_s = \boxed{n_i exp\left(\frac{\phi_{fp}}{V_t}\right)} exp\left(\frac{\Delta\phi_s}{V_t}\right)$$

n_{st}: surface charge density at the threshold inversion point

$$n_s = n_{st} exp\left(\frac{\Delta\phi_s}{V_t}\right)$$



Electron concentration at the surface is the same as the hole concentration in the bulk material. ($n_s = p_b \approx N_a$)

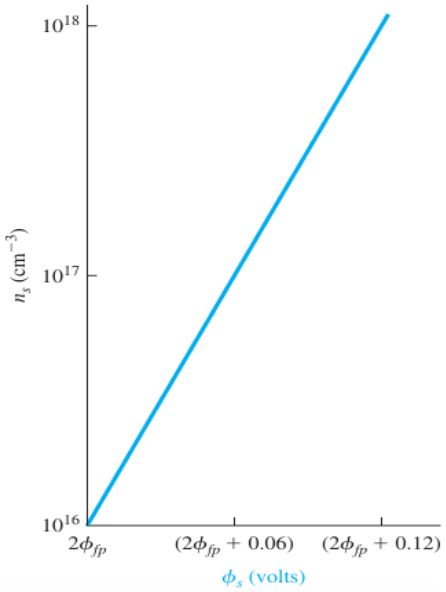
$$n_s = n_i exp\left(\frac{E_F - E_{Fi}}{kT}\right)$$
$$= n_i exp\left(\frac{e\phi_s - e\phi_{fp}}{kT}\right)$$
$$p_s = n_i exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$
$$= n_i exp\left(\frac{e\phi_{fp}}{kT}\right)$$

$n_s = p_b$

$\Rightarrow$ $\phi_S = 2\phi_{fp}$

$\phi_S = 2\phi_{fn}$

(1)The electron inversion charge density increases rapidly with small increases in surface potential.
(2)The space charge width essentially reaches a maximum value.



10.1.4Work Function Differences

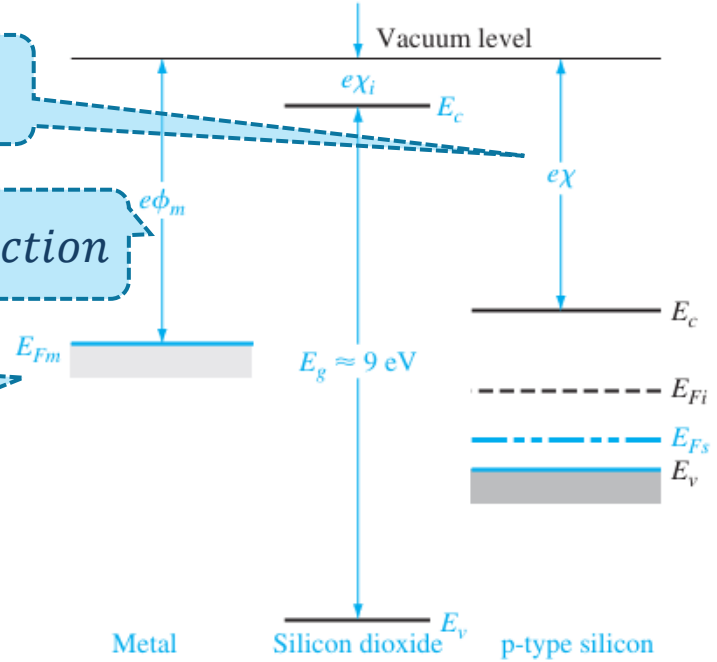
1.Metal gate（p-type substrate)

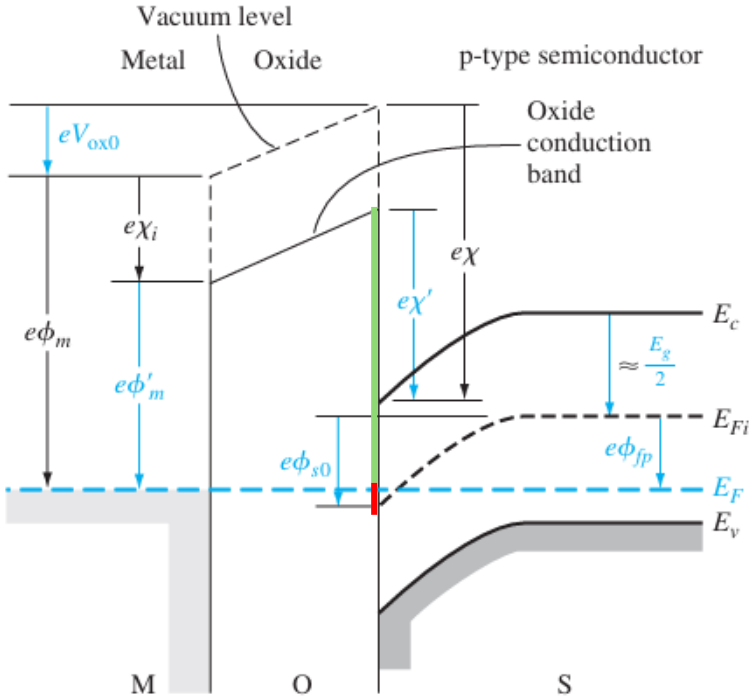
$$\begin{cases} e\phi'_m + eV_{ox0} = e\chi' + \frac{E_g}{2} - e\phi_{s0} + e\phi_{fp} \\ V_{ox0} + \phi_{s0} = -\left[\phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right] \\ \phi_{ms} = \left[\phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right] \end{cases}$$

χ: *electron affinity*

ϕ_m: *metal work function*

$\phi'_m < \phi'_s$

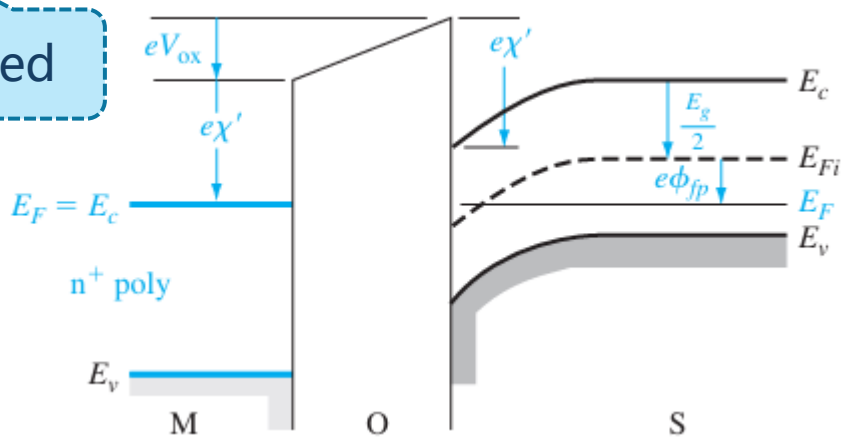


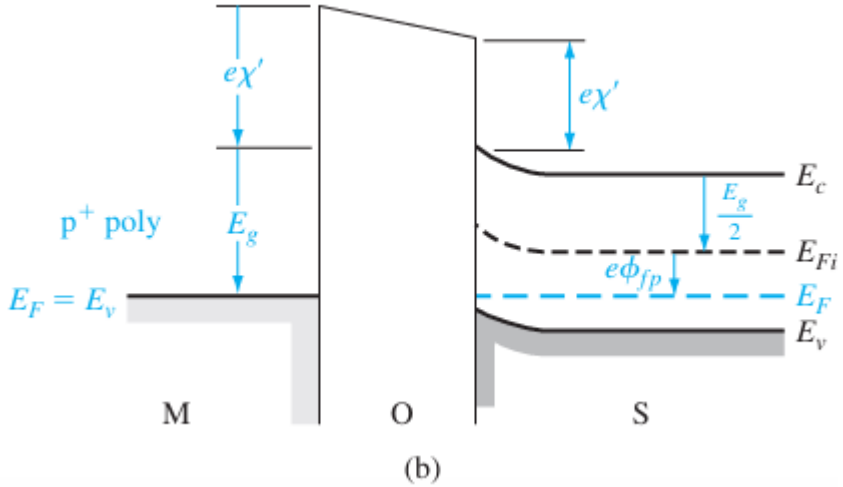


10.1.4Work Function Differences

2.polysilicon gate (p-type substrate)

Degenerately Doped





(1)n+ polysilicon: $\phi_{ms} = \left[\chi' - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right] = -\left(\frac{E_g}{2e} + \phi_{fp}\right)$

(2)p+ polysilicon: $\phi_{ms} = \left[\left(\chi' + \frac{E_g}{e}\right) - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right] = \left(\frac{E_g}{2e} - \phi_{fp}\right)$

10.1.4Work Function Differences

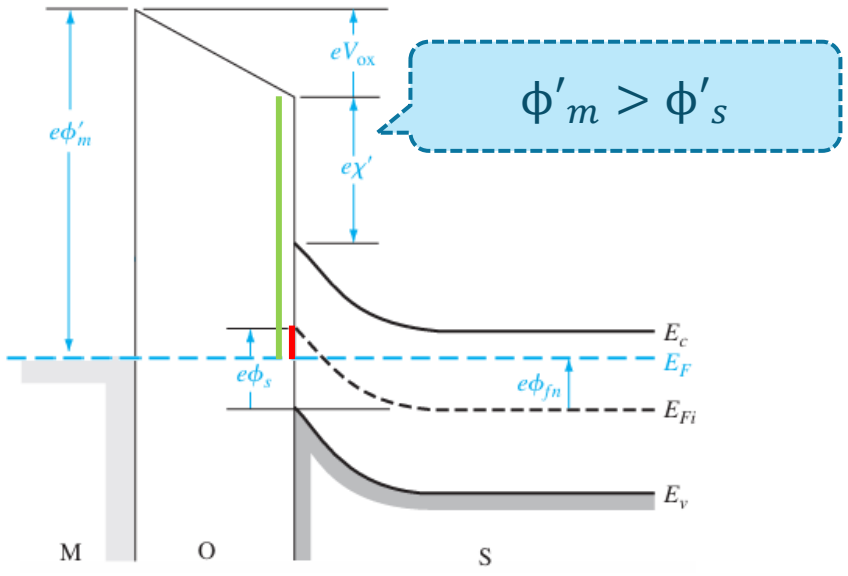
n-type substrate

$$e\phi'_m - eV_{ox0} = e\chi' + \frac{E_g}{2} + e\phi_{s0} - e\phi_{fn}$$

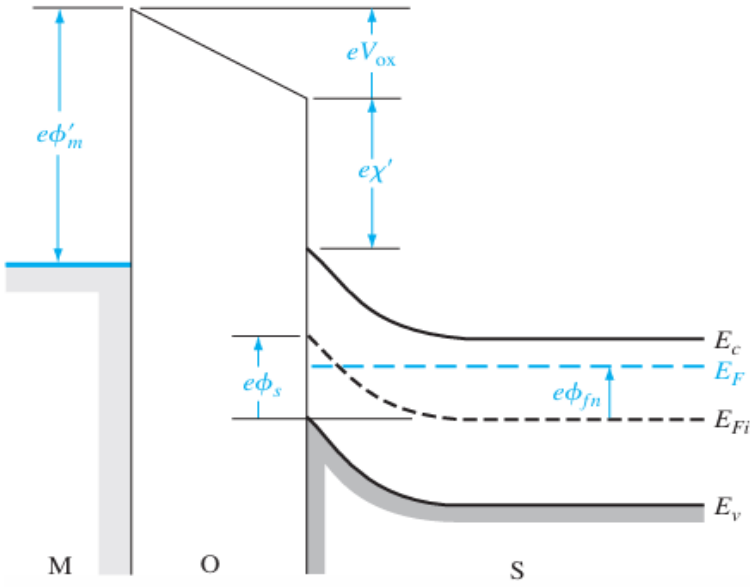
$$V_{ox0} + \phi_{s0} = \left[\phi'_m - \left(\chi' + \frac{E_g}{2e} - \phi_{fn}\right)\right]$$

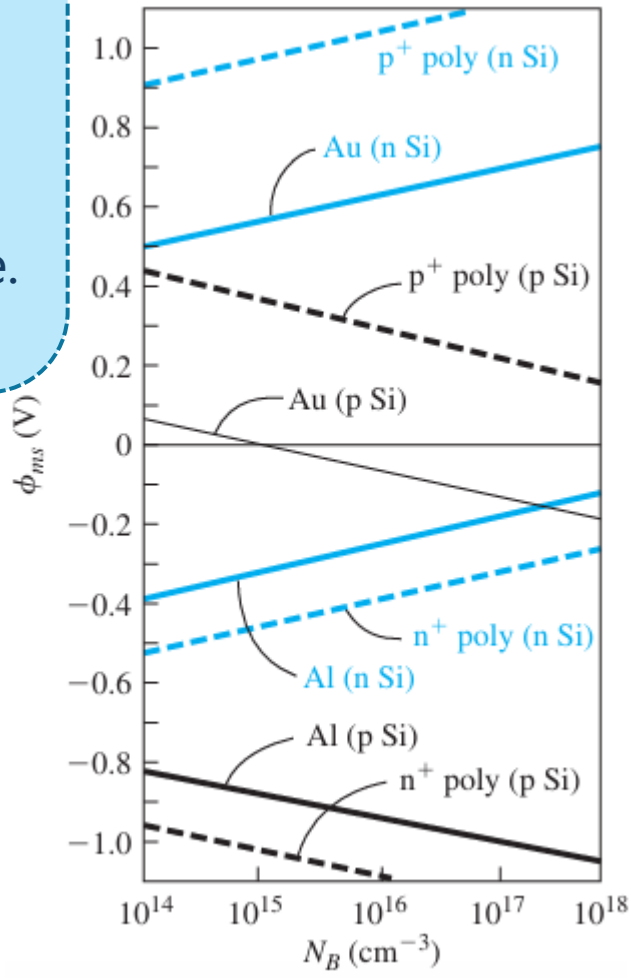
$$\phi_{ms} = \left[\phi'_m - \left(\chi' + \frac{E_g}{2e} - \phi_{fn}\right)\right]$$

The fermi level is not equal to the conduction-band energy for the n+ gate and is not equal to the valence-band energy for the p+ gate.



$\phi'_m > \phi'_s$





10.1.5 Flat-band Voltage

Flat-band voltage (V_{FB}) is defined as the applied gate voltage such that there is no band bending in the semiconductor and ,as a result, zero net space charge in this region.

$V_{FB} \neq 0$ $\begin{cases} \text{Work function difference} \\ \text{Possible trapped charge in the oxide} \end{cases}$

$$V_{ox0} + \phi_{s0} = -\phi_{ms}$$

$$V_G = \Delta V_{ox0} + \Delta\phi_s = (V_{ox} - V_{ox0}) + (\phi_s - \phi_{s0})$$

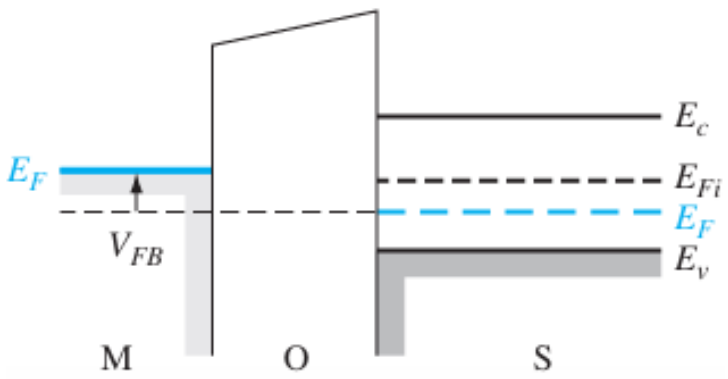
$$V_G = V_{ox} + \phi_s + \phi_{ms}$$

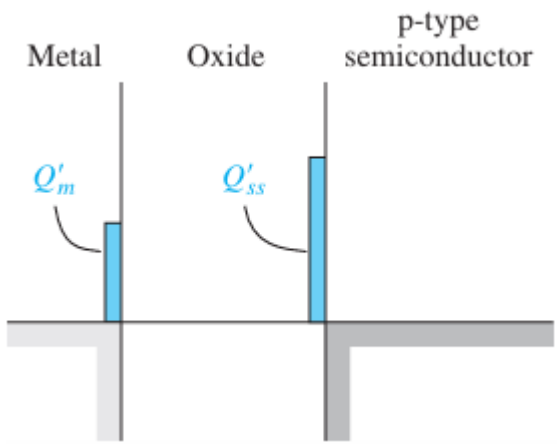
$$V_G = V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

$$Q'_m + Q'_{ss} = 0$$

$$V_{ox} = \frac{Q'_m}{C_{ox}}$$

$$V_{ox} = -\frac{Q'_{ss}}{C_{ox}}$$





10.1.6 Threshold Voltage

The threshold voltage is defined as the applied gate voltage required to achieve the threshold inversion point.

$$Q'_{mT} + Q'_{SS} = |Q'_{SD}(max)|$$

$$Q'_{SD}(max) = eN_a x_{dT}$$

$$V_{oxT} = \frac{Q'_{mT}}{C_{ox}}$$

$$V_{oxT} = \frac{1}{C_{ox}}(|Q'_{SD}(max)| - Q'_{SS})$$

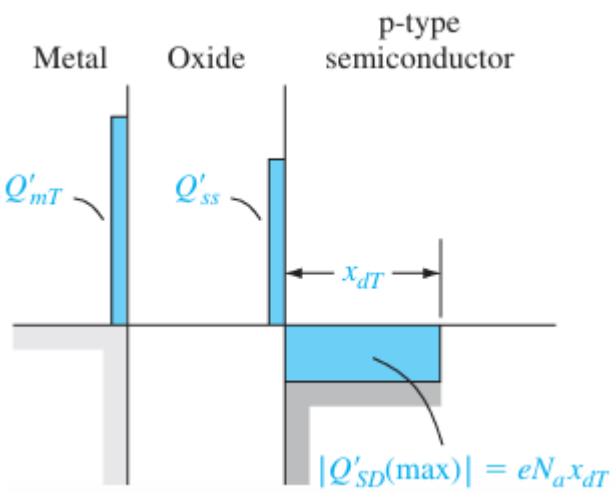
$$V_G = \Delta V_{ox0} + \Delta\phi_s = V_{ox} + \phi_s + \phi_{ms}$$

$$V_{TN} = V_{oxT} + 2\phi_{fp} + \phi_{ms}$$

$$V_{TN} = \frac{|Q'_{SD}(max)|}{C_{ox}} + V_{FB} + 2\phi_{fp}$$

$$V_{TN} = \frac{|Q'_{SD}(max)|}{C_{ox}} \boxed{- \frac{Q'_{SS}}{C_{ox}} + \phi_{ms}} + 2\phi_{fp}$$

$$V_{TN} = (|Q'_{SD}(max)| - Q'_{SS})\frac{t_{ox}}{\epsilon_{ox}} + 2\phi_{fp} + \phi_{ms}$$



Oxide thickness

Oxide charge

Semiconductor doping

Gate metal

Oxide material

Semiconductor material

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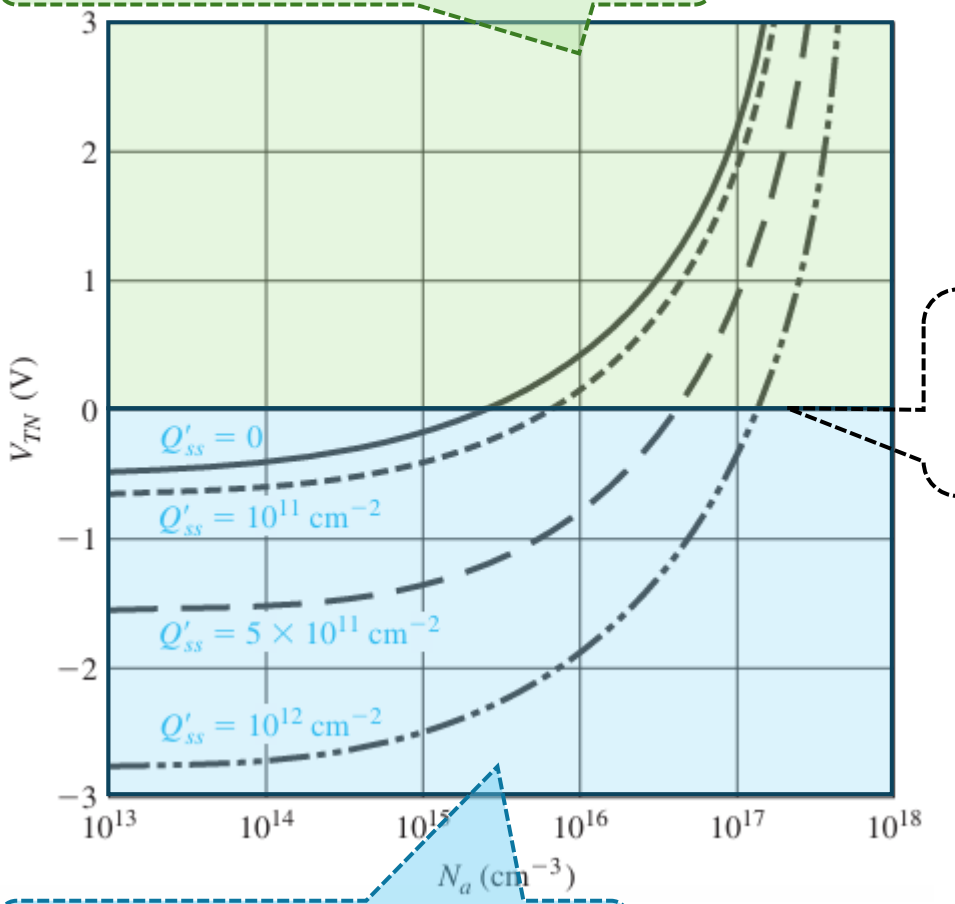
P-type substrate

Enhancement mode($V_{TN} > 0$)



$Q'_{SS} \uparrow$, $|V_{TN}| \uparrow$
$N_a \uparrow$, V_{TN}

Depletion mode($V_{TN} < 0$)

n-type substrate

$$V_{TP}=(-|Q'_{SD}(max)|-Q'_{SS})\frac{t_{ox}}{\epsilon_{ox}}-2\phi_{fp}+\phi_{ms}$$

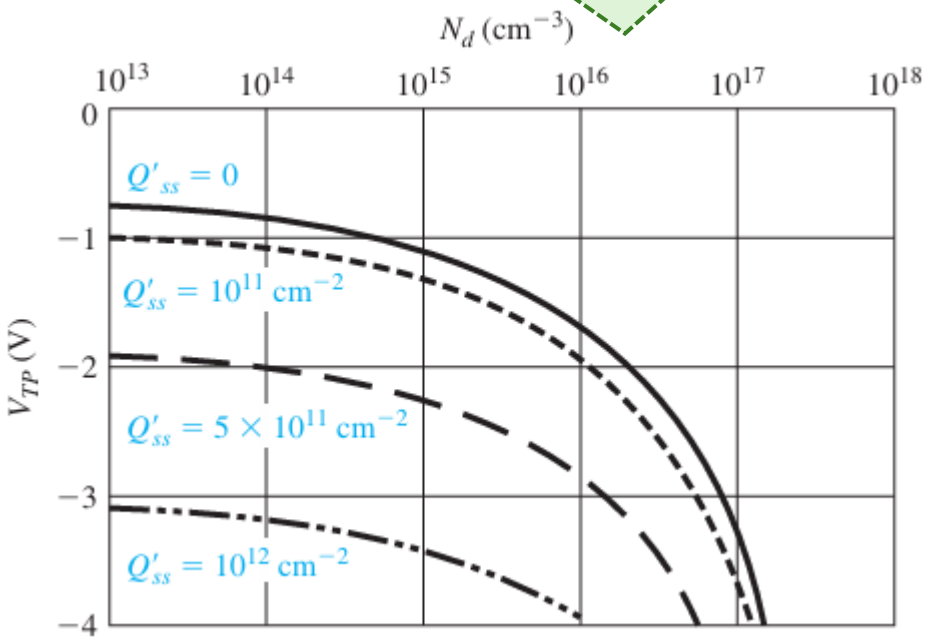
$$\phi_{ms}=\left[\phi'_m-\left(\chi'+\frac{E_g}{2e}-\phi_{fn}\right)\right]$$

$$Q'_{SD}(max)=eN_d x_{dT}$$

$$x_{dT}=\left(\frac{4\epsilon_s\phi_{fn}}{eN_d}\right)^{1/2}$$

$$\phi_{fn}=V_t ln\left(\frac{N_d}{n_i}\right)$$

Enhancement mode($V_{Tp}<0$)

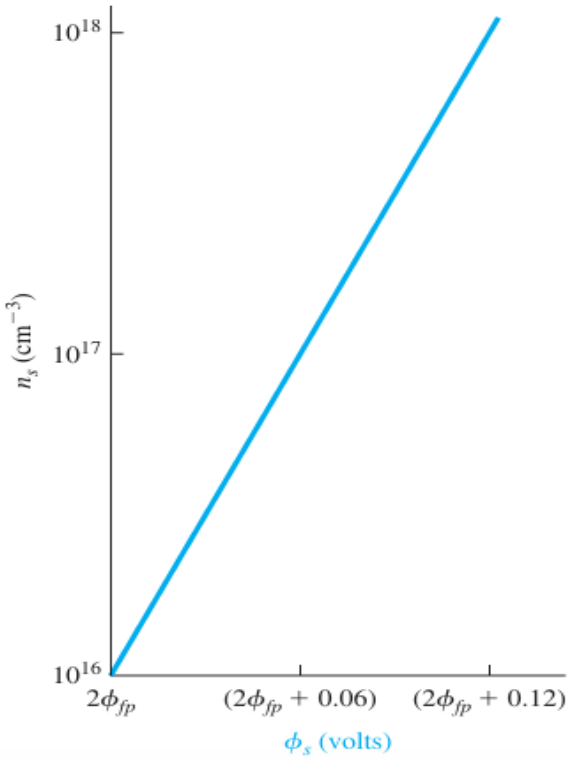


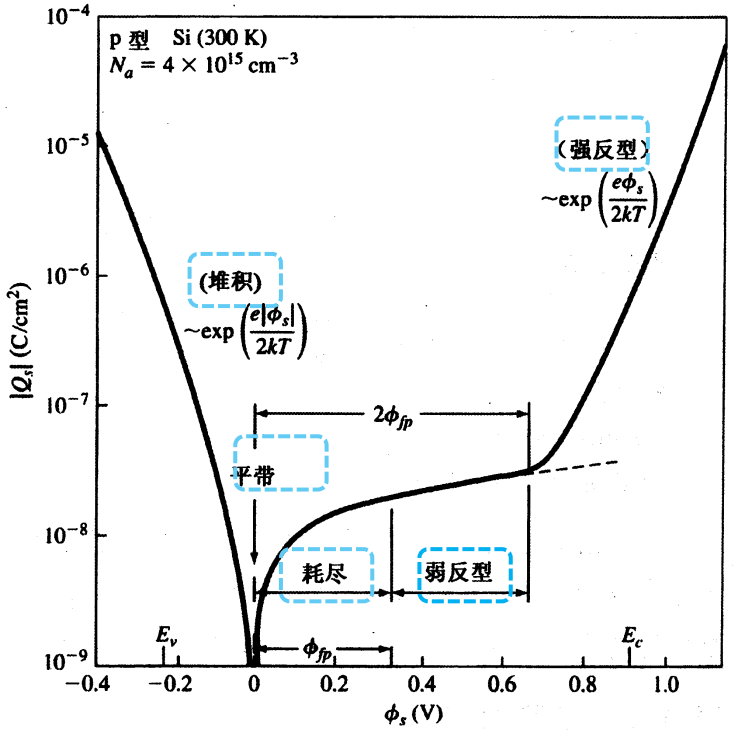
	MOS with P-type substrate	MOS with N-type substre
Fermi potential	$\phi_{fp} = V_t ln\left(\frac{N_a}{n_i}\right)$	$\phi_{fn} = V_t ln\left(\frac{N_d}{n_i}\right)$
Maximum space charge width	$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$	$x_{dT} = \left(\frac{4\epsilon_s \phi_{fn}}{eN_d}\right)^{1/2}$
Maximum space charge density per unit area of the depletion region	$\lvert Q_{SD}'_{\max} \rvert = eN_a x_{dT}$	$\lvert Q_{SD}'_{\max} \rvert = eN_d x_{dT}$
Oxide capacitance per unit area	$C_{ox} = \varepsilon_{ox} / t_{ox}$	$C_{ox} = \varepsilon_{ox} / t_{ox}$
Flat-band voltage	$V_{FB} = \phi_{ms} - Q'_{ss} / C_{ox}$	$V_{FB} = \phi_{ms} - Q'_{ss} / C_{ox}$
Work function difference	$\phi_{ms} = \phi_m' - (\chi' + \frac{E_g}{2e} + \phi_{fp})$	$\phi_{ms} = \phi_m' - (\chi' + \frac{E_g}{2e} - \phi_{fn})$
Threshold voltage	$V_{TN} = \frac{\lvert Q'_{SD\max} \rvert}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp}$ $V_{TN} = 0.5 \sim 0.7\text{V}$	$V_{TP} = -\frac{\lvert Q'_{SD\max} \rvert}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} - 2\phi_{fn}$ $V_{TP} = -0.4 \sim -0.7\text{V}$

10.2 CAPACITANCE–VOLTAGE CHARACTERISTICS

Relationship between the surface charge density and the surface potential

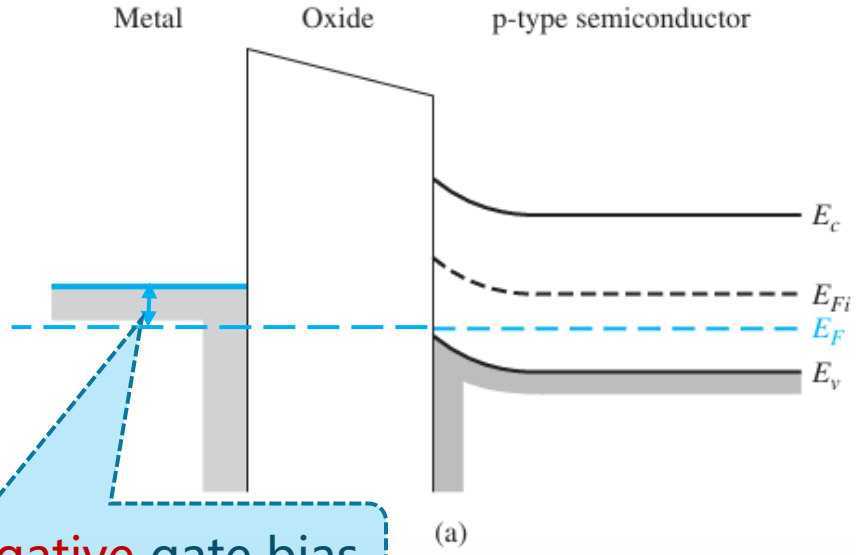
$$C = \frac{dQ}{dV}$$



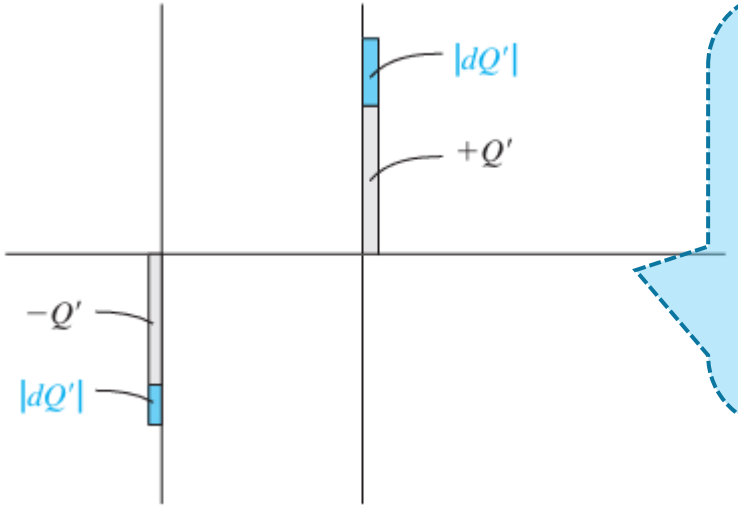


10.2.1 Ideal C–V Characteristics

1.Accumulation mode

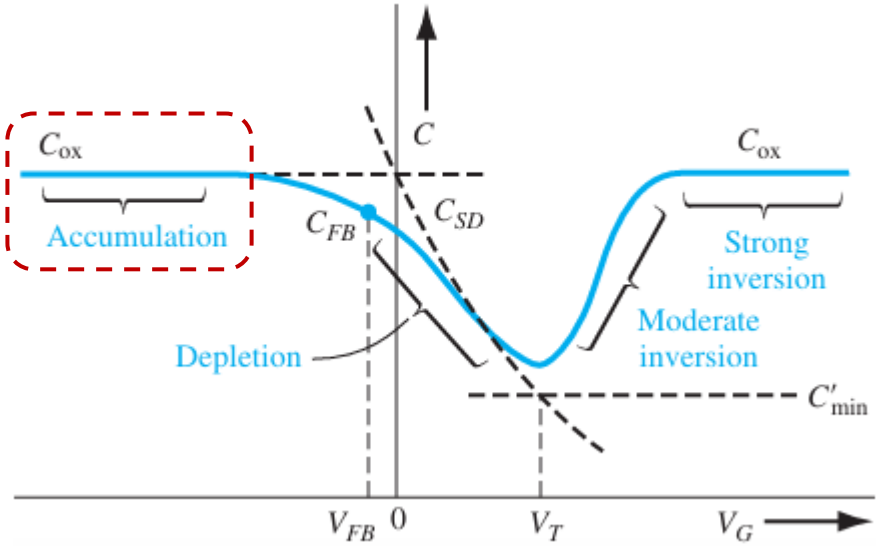


negative gate bias



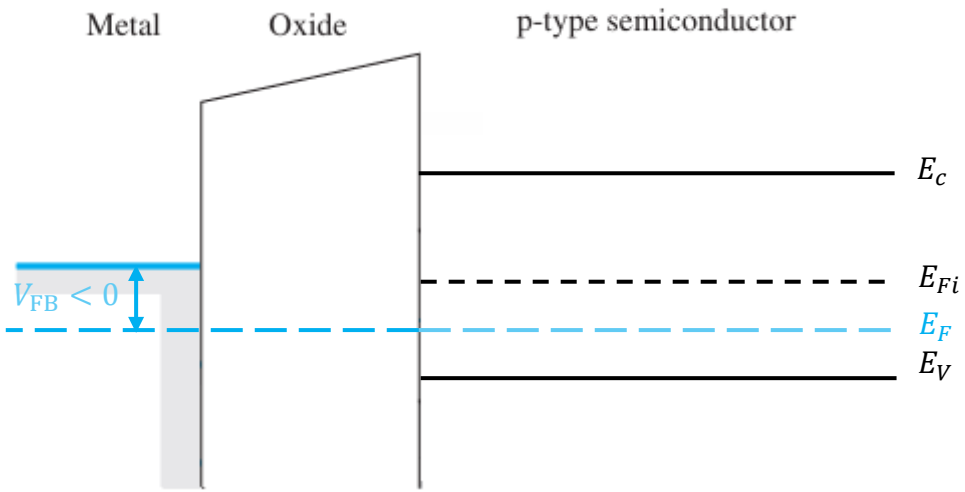
As a parallel-plate capacitor:

$$C'(acc) = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$



10.2.1 Ideal C–V Characteristics

2.flat-band mode



$$C'_{FB} = \frac{\epsilon_{ox}}{t_{ox} + \frac{\epsilon_{ox}}{t_{ox}}\sqrt{\frac{kT}{e}\frac{\epsilon_s}{eN_a}}}$$

10.2.1 Ideal C–V Characteristics

3.depletion mode

$$\frac{1}{C'(depl)} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}}$$

$$C'(depl) = \frac{C_{ox}C'_{SD}}{C_{ox} + C'_{SD}}$$

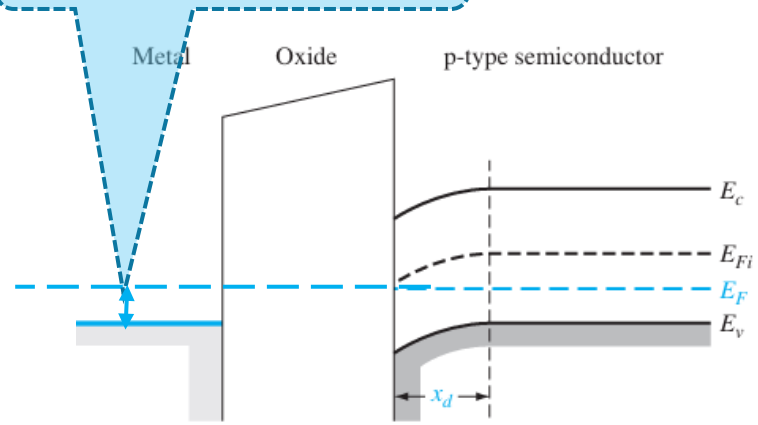
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

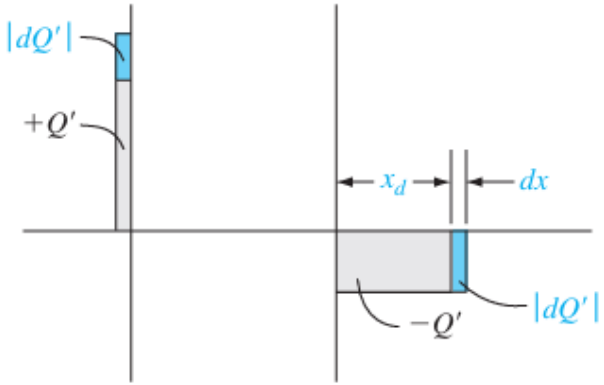
$$C'_{SD} = \frac{\epsilon_s}{x_d}$$

$$C'(depl) = \frac{C_{ox}}{1+\frac{C_{ox}}{C'_{SD}}} = \frac{\epsilon_{ox}}{t_{ox}+(\frac{\epsilon_{ox}}{\epsilon_s})x_d}$$

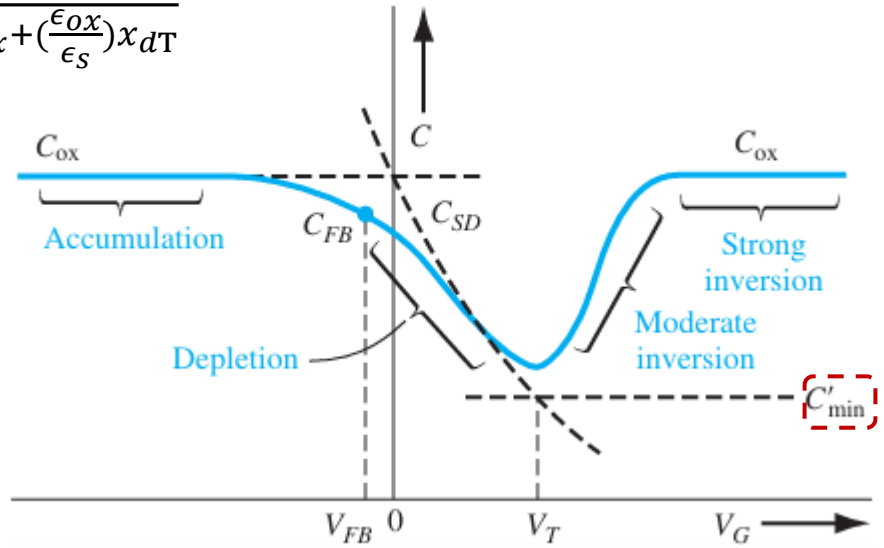
positive gate bias

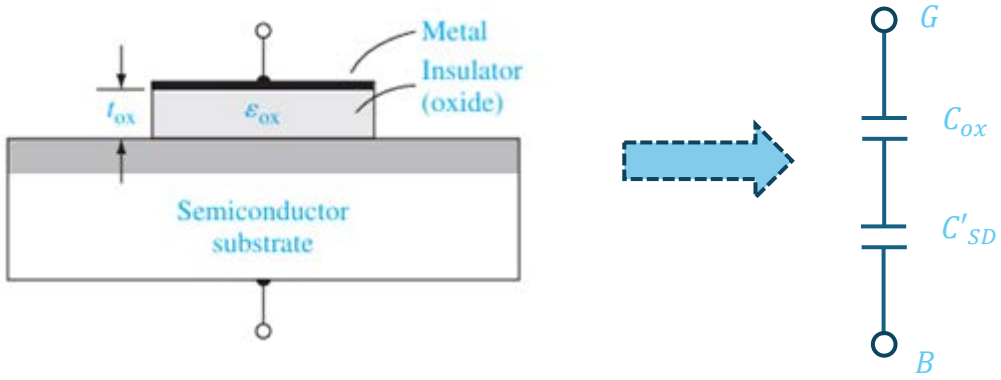
$x_d \uparrow \Rightarrow C'(depl) \downarrow$





$$C'_{min} = \frac{C_{ox}}{1+\frac{C_{ox}}{C'_{SD}}} = \frac{\epsilon_{ox}}{t_{ox}+(\frac{\epsilon_{ox}}{\epsilon_s})x_{dT}}$$

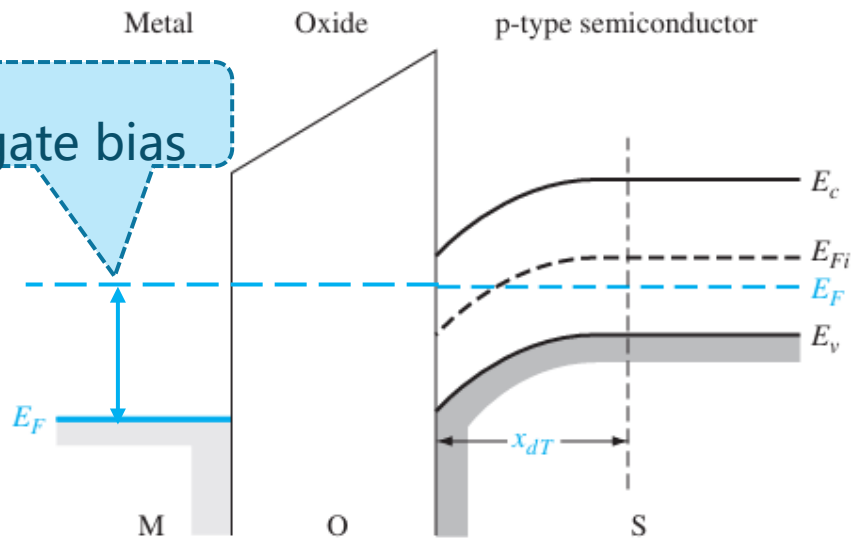




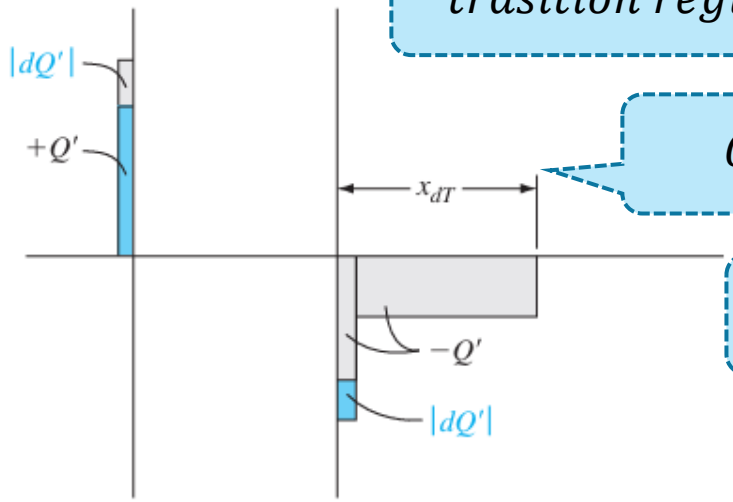
10.2.1 Ideal C–V Characteristics

4.inversion mode(strong inversion and Moderate inversion)

" large "
positive gate bias

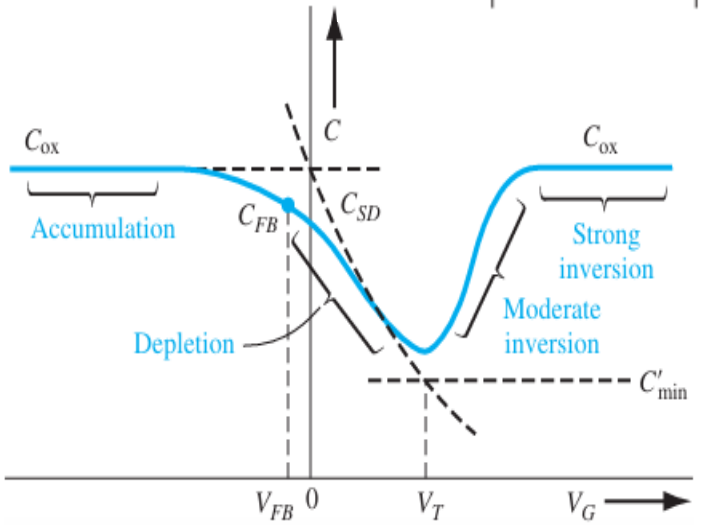


trasition region



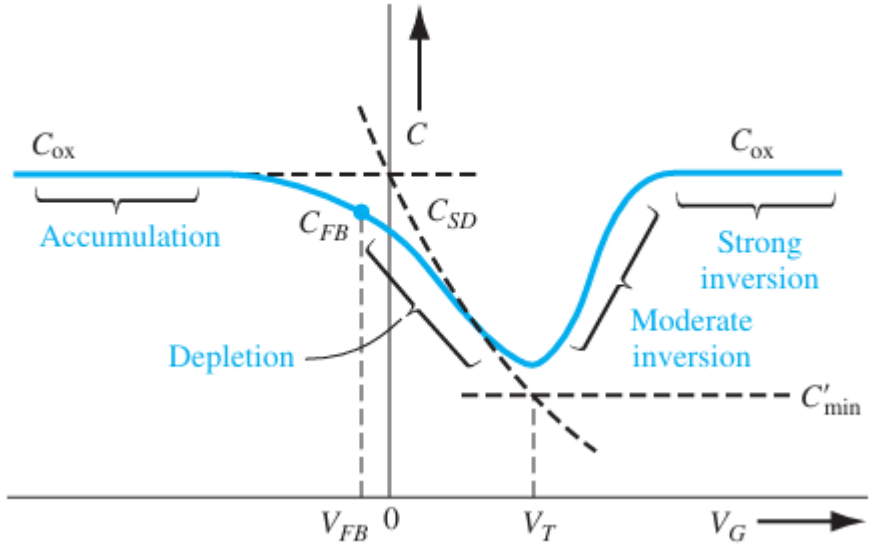
$$C'(inv) = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

1.Strong inversion
2.low frequency

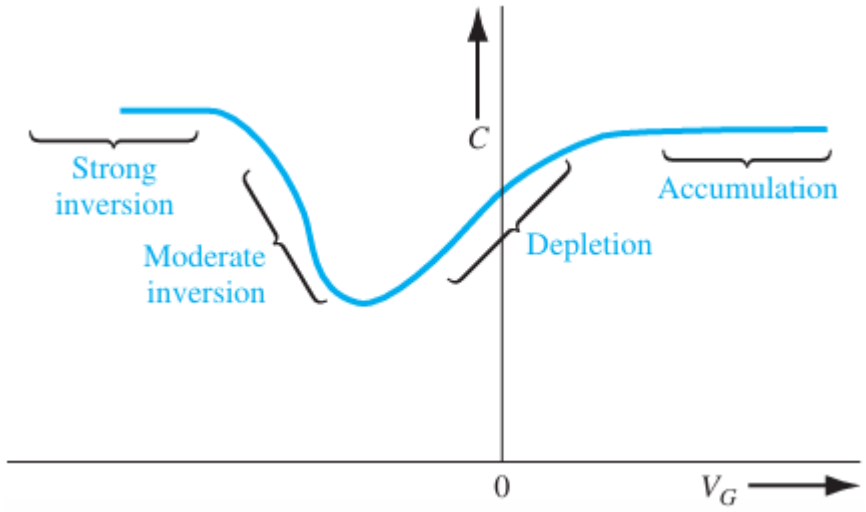


10.2.1 Ideal C–V Characteristics

MOS capacitor with a p-type substrate



MOS capacitor with a n-type substrate



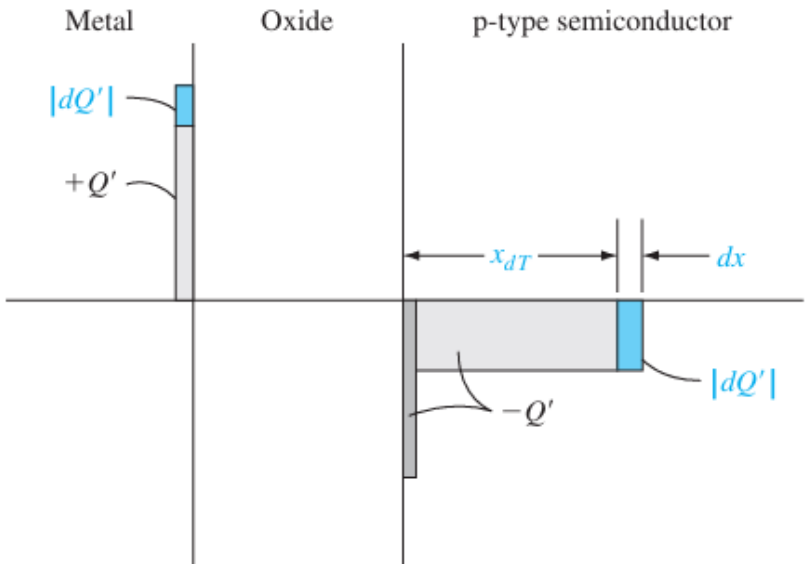
10.2.2 Frequency Effects

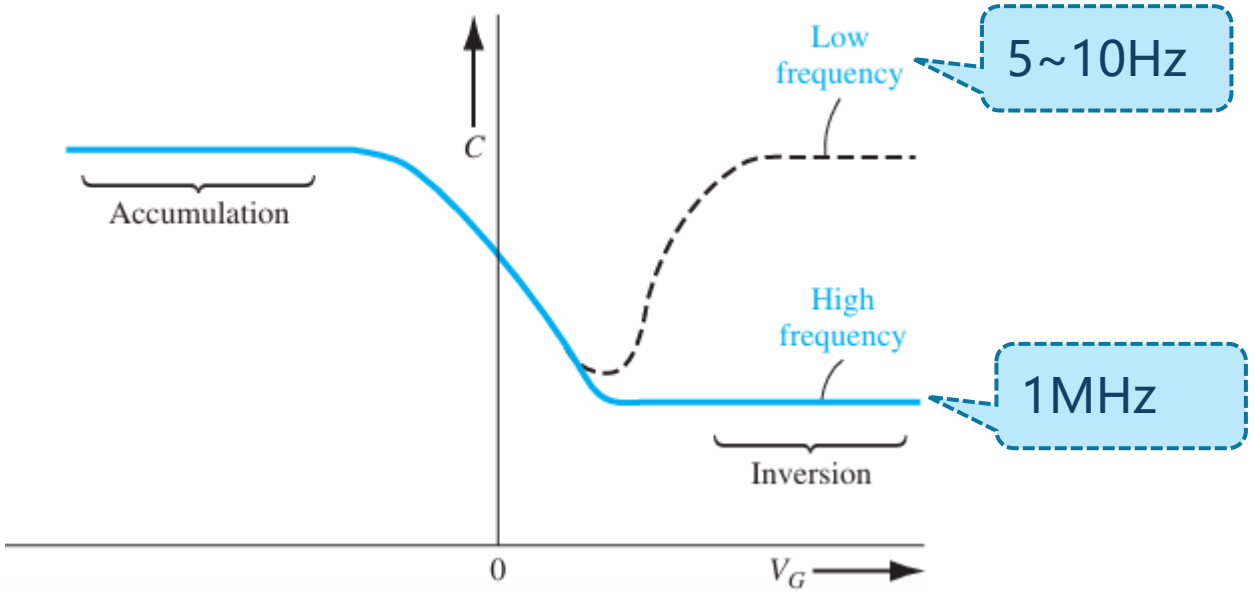
The source of electrons in the inversion layer

diffusion of minority carrier electrons from the p-type substrate across the space charge region

Thermal generation of electron-hole pairs within the space charge region

Electron concentration can not change instantaneously.





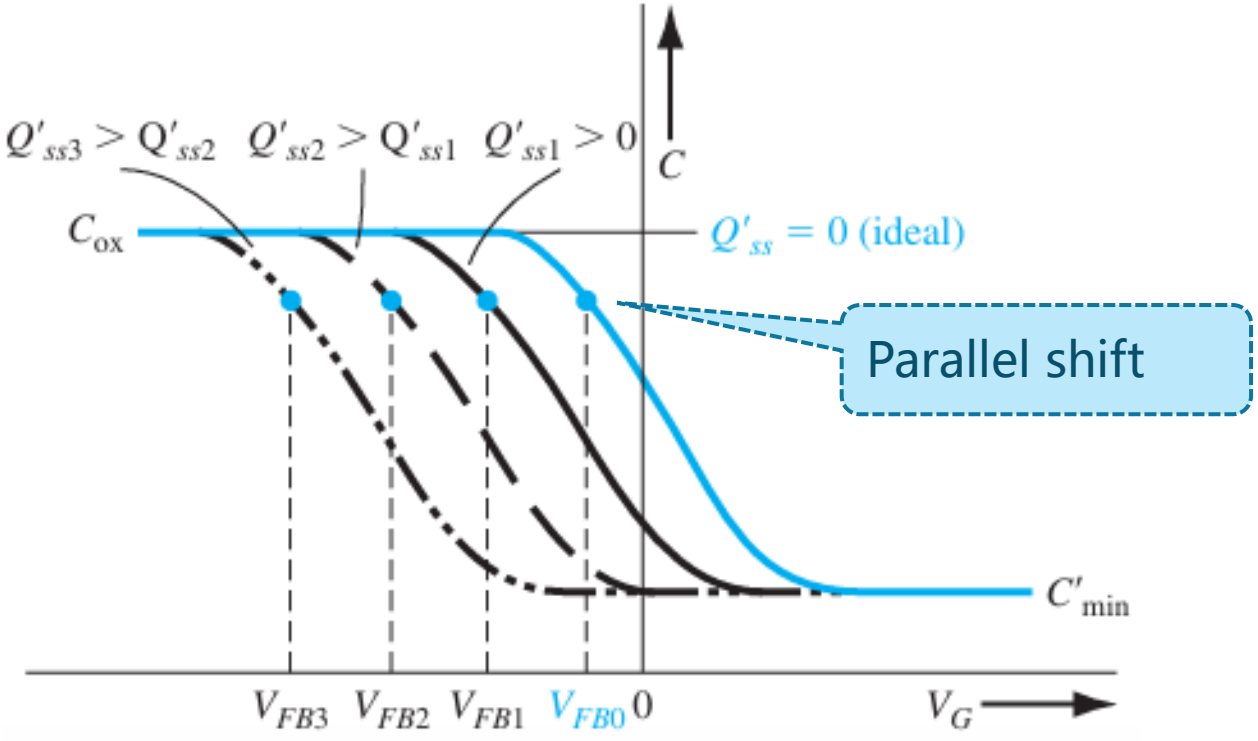
10.2.3 Fixed Oxide and Interface Charge Effects

1. fixed oxide charge

positive

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

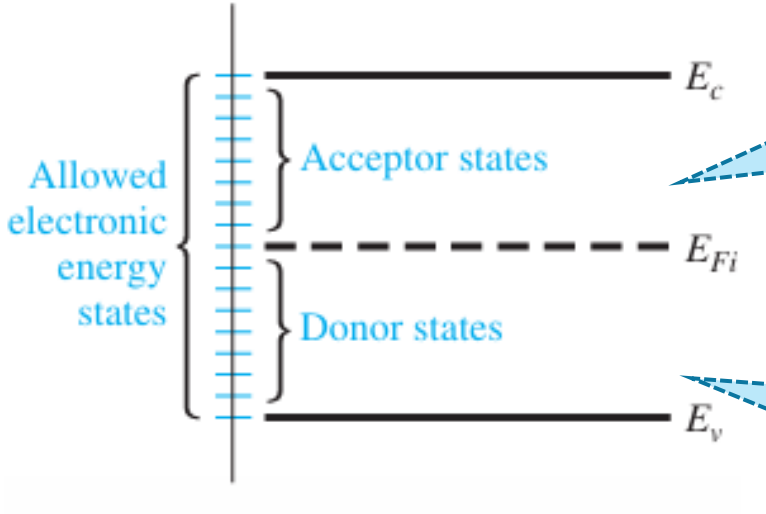
The flat-band voltage shifts to more negative voltages for a positive fixed oxide charge. $Q'_{ss} \uparrow \Rightarrow V_{FB} \rightarrow -$



Parallel shift

10.2.3 Fixed Oxide and Interface Charge Effects

2. Interface states: The periodic nature of the semiconductor is abruptly terminated at the interface so that the allowed electronic energy levels will exist within the forbidden bandgap



An acceptor state is neutral if the Fermi level is below the state and becomes negatively charged if the Fermi level is above the state.

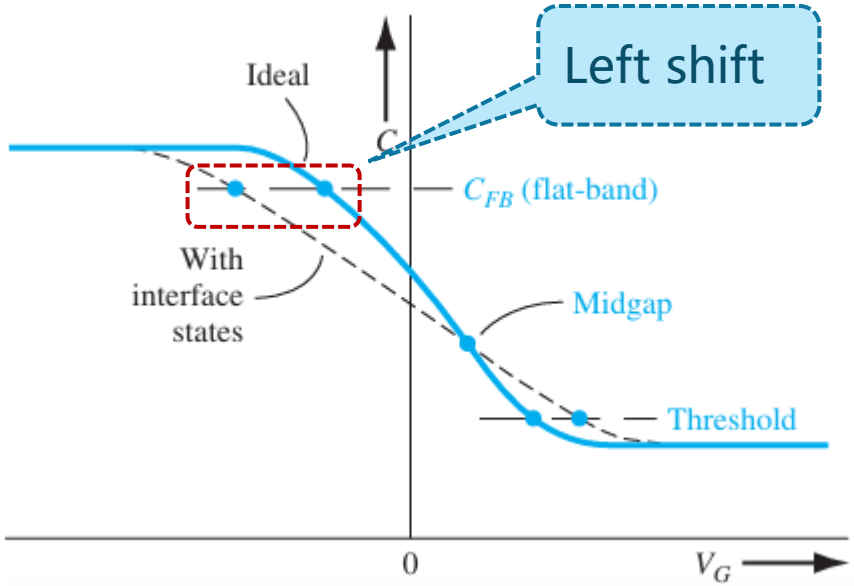
A donor state is neutral if the Fermi level is above the state and becomes positively charged if the Fermi level is below the state.

10.2.3 Fixed Oxide and Interface Charge Effects

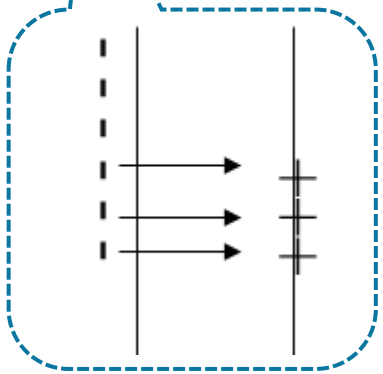
(1) Accumulation mode

a p-type semiconductor of a MOS capacitor

More negative charges are needed on the metal gate to neutralize the positive interface charges, so the flat band voltage becomes more negative.

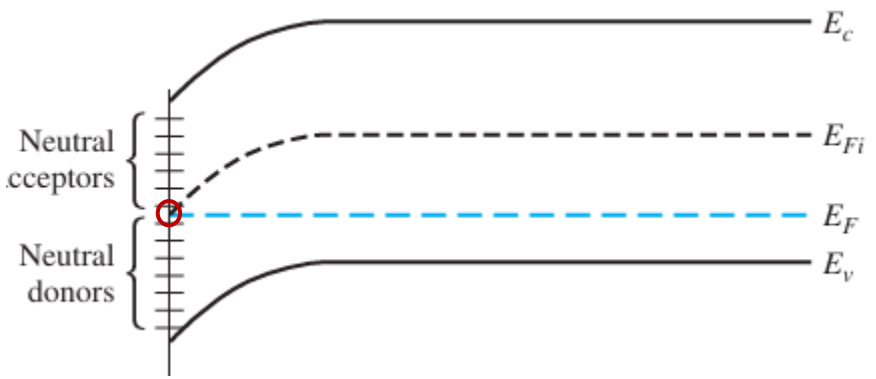


Left shift

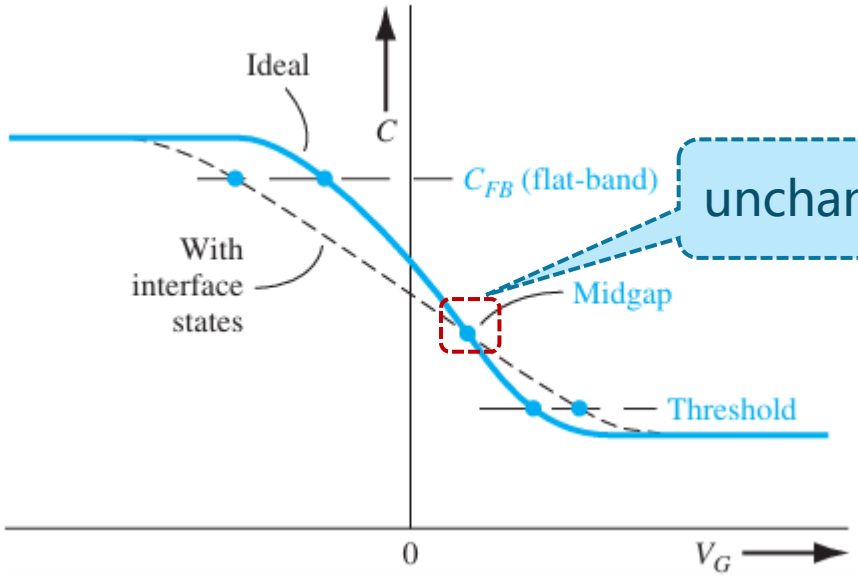


10.2.3 Fixed Oxide and Interface Charge Effects

(2) Midgap



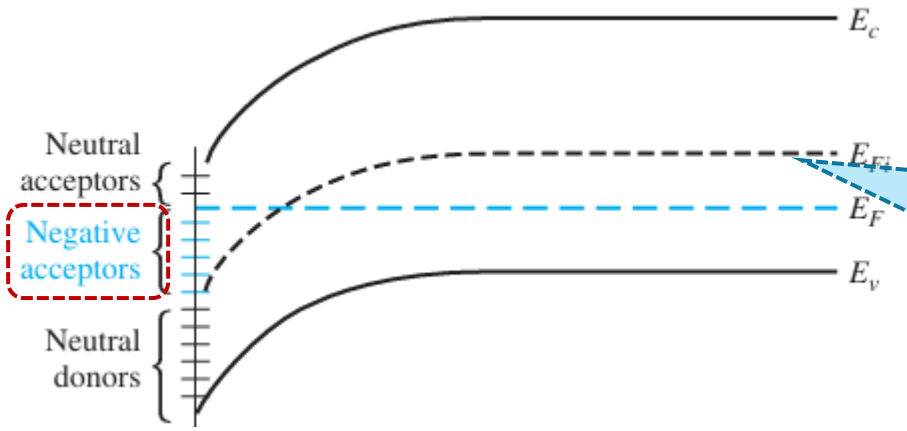
All interface states are neutral



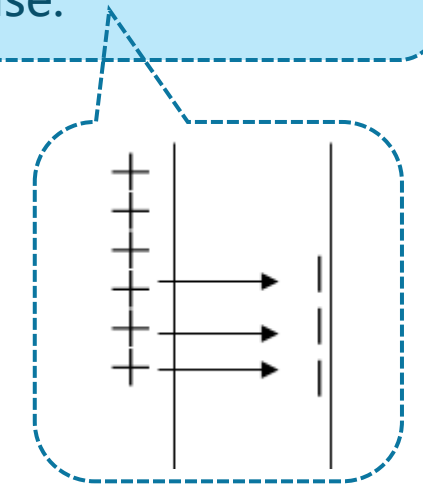
unchanged

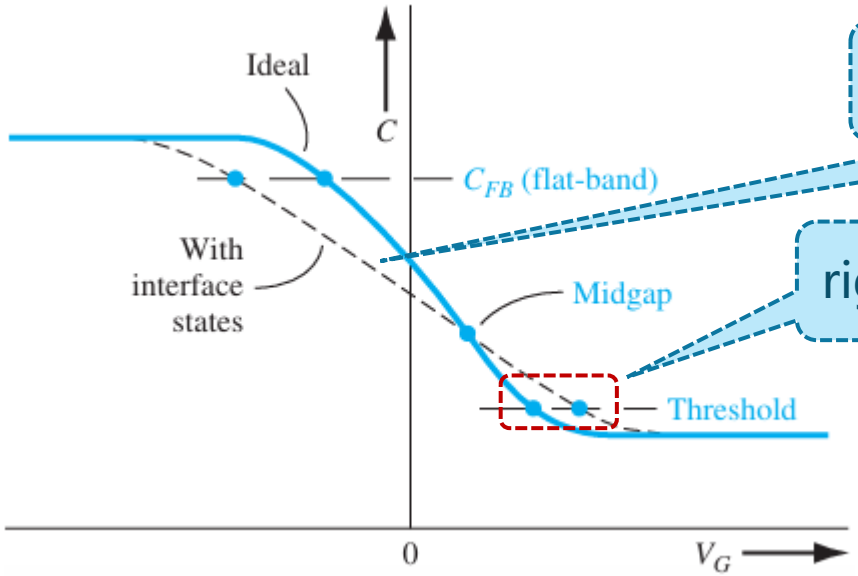
10.2.3 Fixed Oxide and Interface Charge Effects

(3) Inversion mode



More positive charges are needed on the metal gate to neutralize the negative interface charges, so the threshold voltage increase.





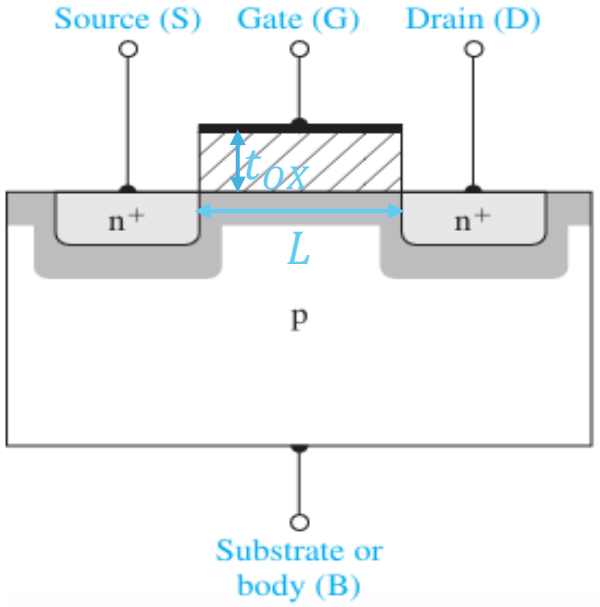
The C-V cures become "smeared out"

right shift

10.3 | THE BASIC MOSFET OPERATION
10.3.1 MOSFET Structures

1.structure

- Channel length (L)
- Channel width (W)
- Oxide thickness (t_{OX})

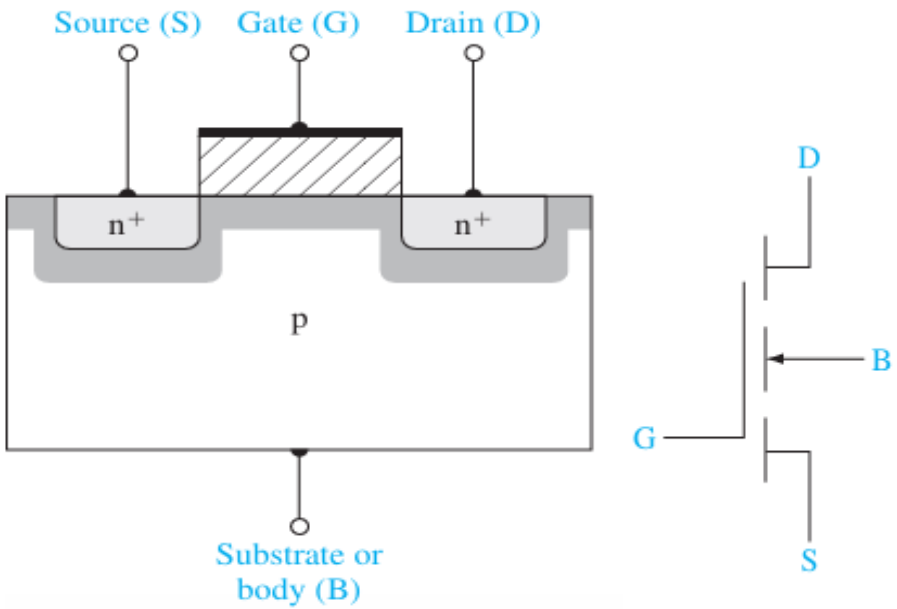


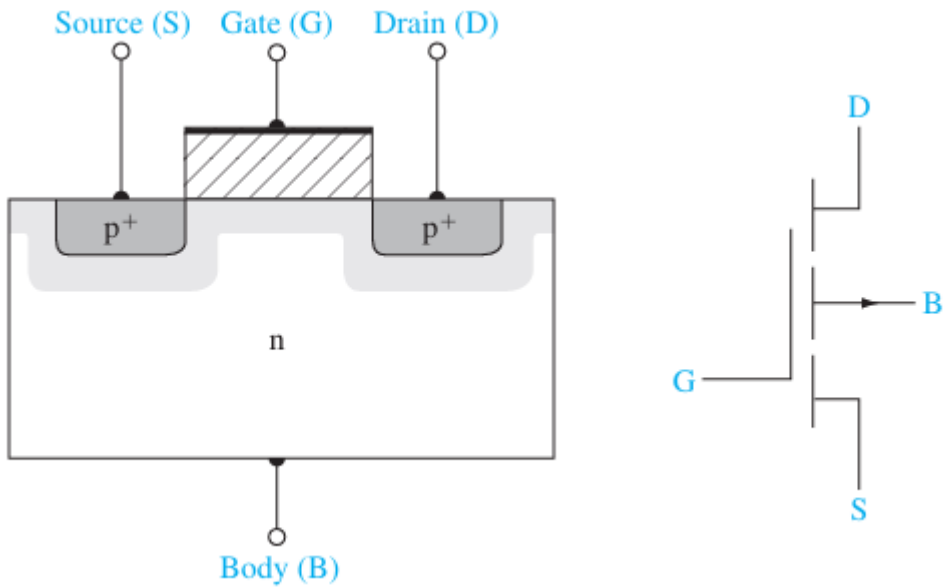
2. classification

- Conductivity type of the channel
  - n-channel MOSFET
  - p-channel MOSFET
- depletion mode MOSFET — Transistor is ON when $V_g = 0$
- enhancement mode MOSFET — Transistor is OFF when $V_g = 0$

10.3.1 MOSFET Structures

2. classification——n-channel vs. p-channel





- P-type substrate
- n-type channel
- $V_{\mathrm{DS}} > 0$
- Electrons flow from source to the drain terminal

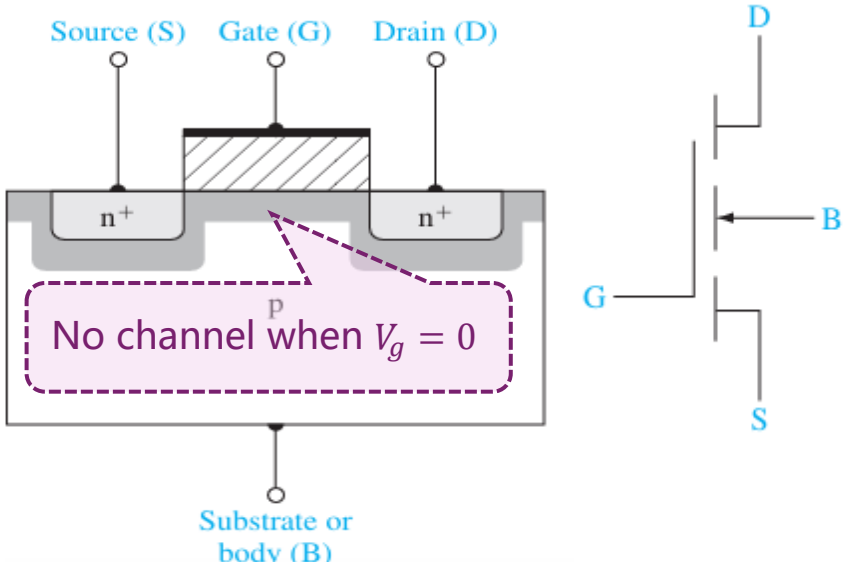
- n-type substrate
- p-type channel
- $V_{\mathrm{DS}} < 0$
- Holes flow from source to the drain terminal

10.3.1 MOSFET Structures

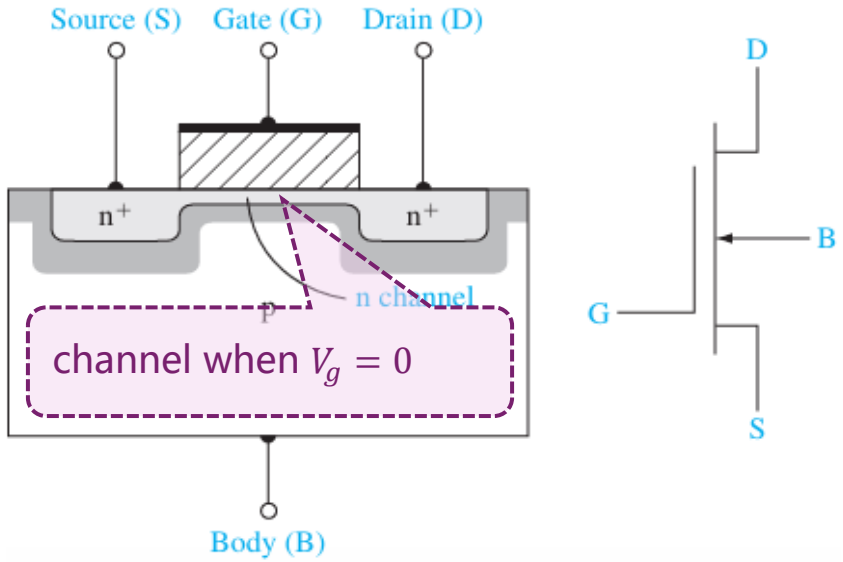
normally OFF

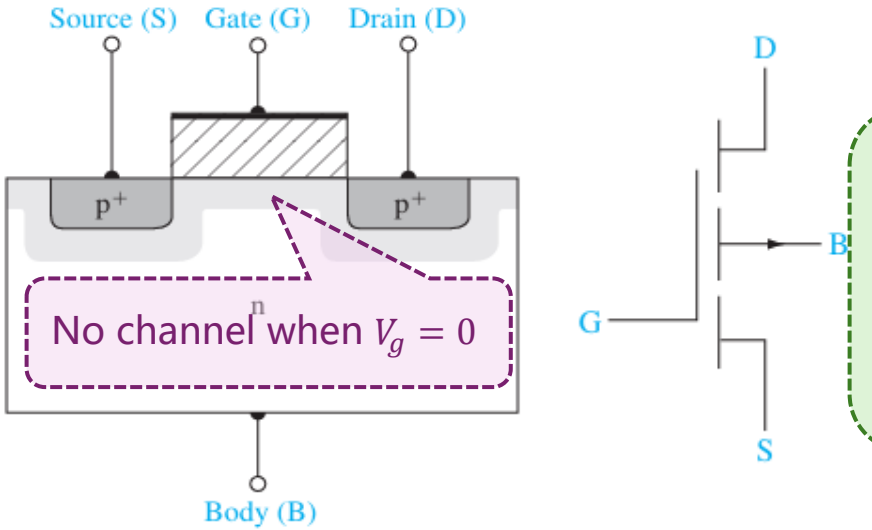
normally ON

2. classification——Enhancement Mode vs. Depletion Mode

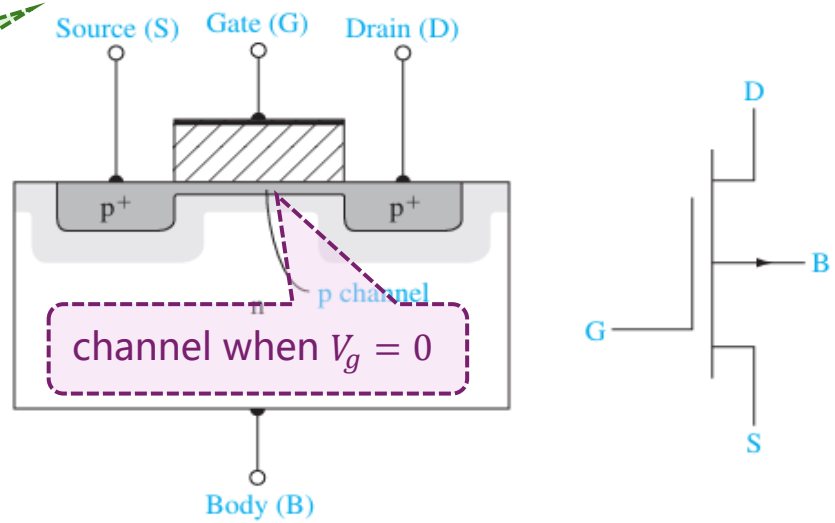


Conduction between source and drain regions is *enhanced* by applying a gate voltage

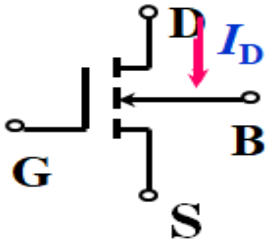
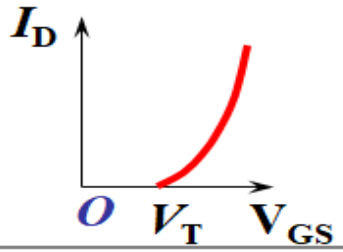
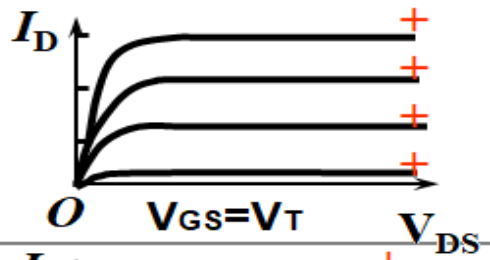
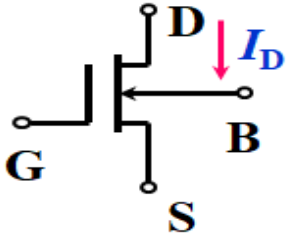
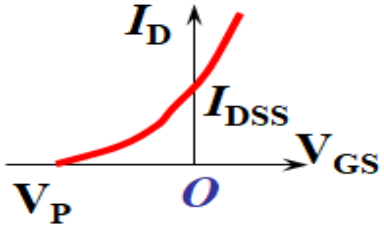
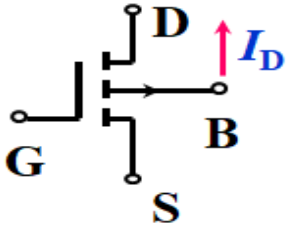
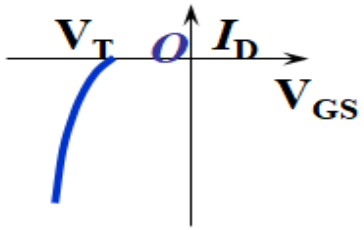
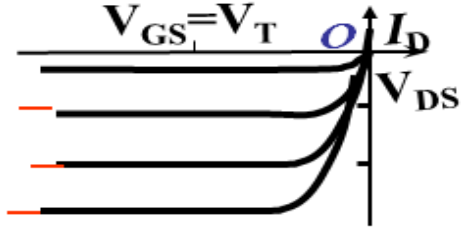
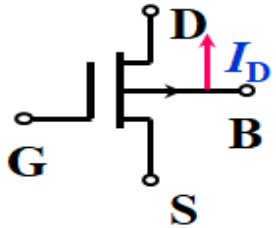
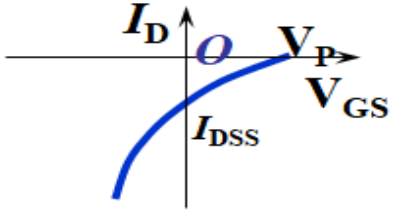
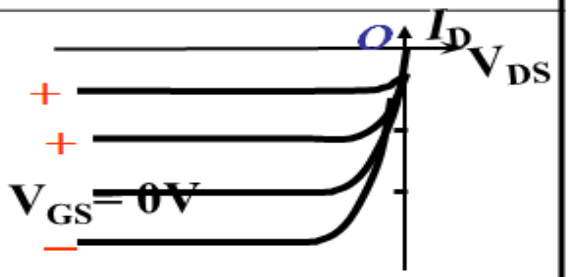




A gate voltage must be applied to deplete the channel region in order to turn off the transistor



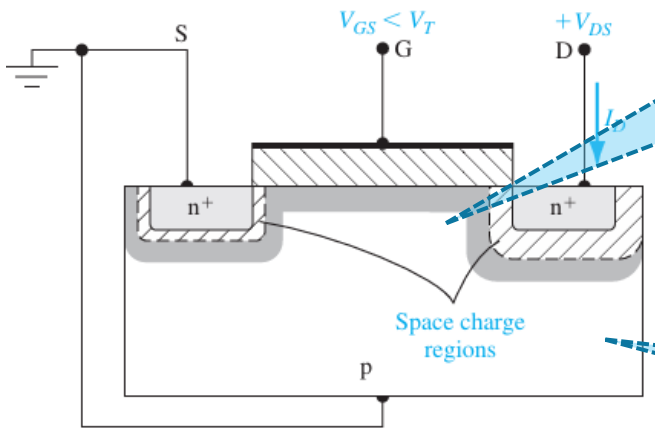
10.3.1 MOSFET Structures

classification		Circuit symbol	Transfer Characteristic	Output Characteristic
NMOS	Enhancement mode			
	depletion mode			
PMOS	Enhancement mode			
	depletion mode			

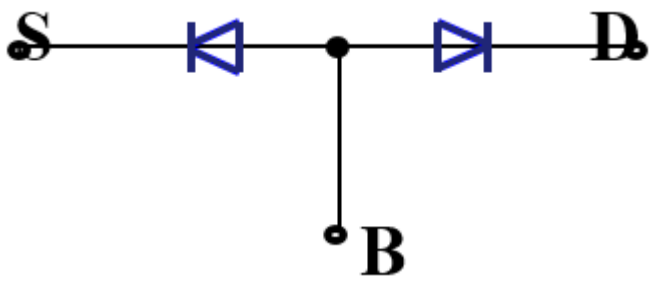
10.3.2 Current–Voltage Relationship—Concepts

1.The influence of V_{GS}

(1)$V_{GS} = 0$



The connection between the drain and source is equivalent to two back-to-back PN junctions.



MOSFET is OFF

(2)$V_{GS} > 0$

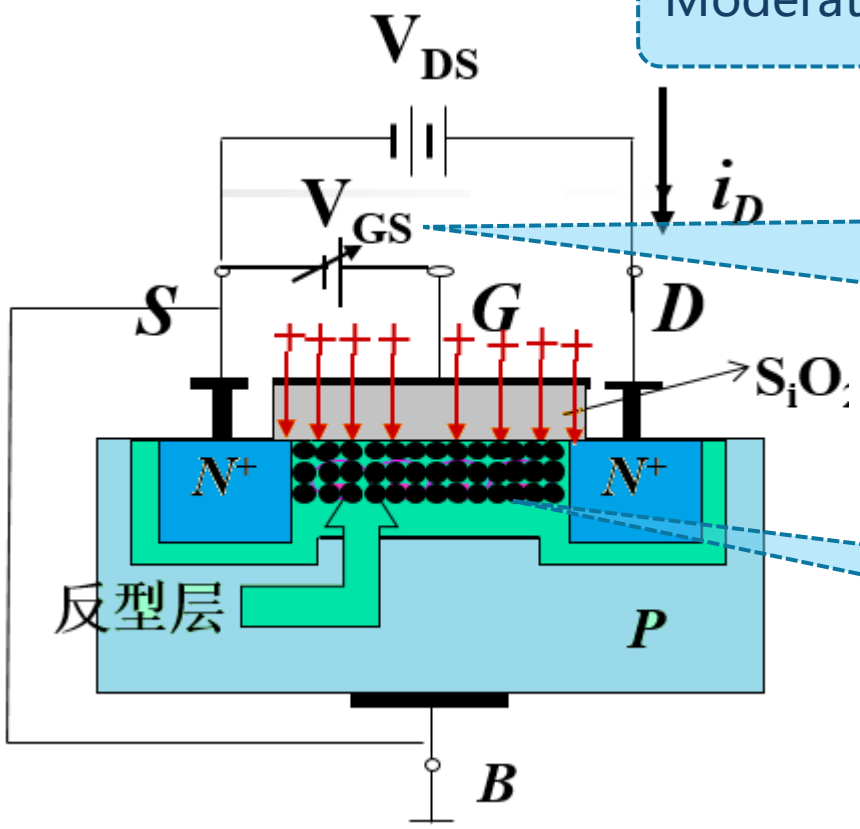
Holes are repelled from the oxide-semiconductor interface.

As V_{GS} increases,the width of the depletion layer increases.

10.3.2 Current–Voltage Relationship—Concepts

(3) V_{GS} increases to V_T

Moderate inversion ⇨ strong inversion



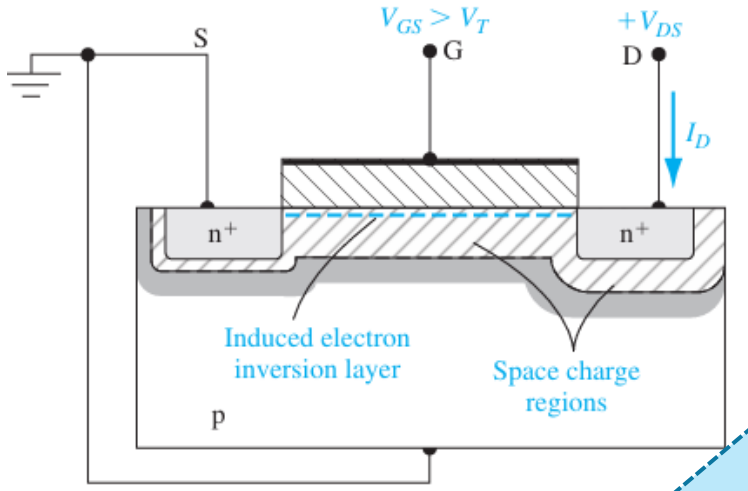
As V_{GS} increases, the channel thickens and the channel resistance decreases, Under a certain V_{DS}, I_{DS} will increase.

The inversion layer (channel)

10.3.2 Current–Voltage Relationship—Concepts

2.The influence of V_{DS}

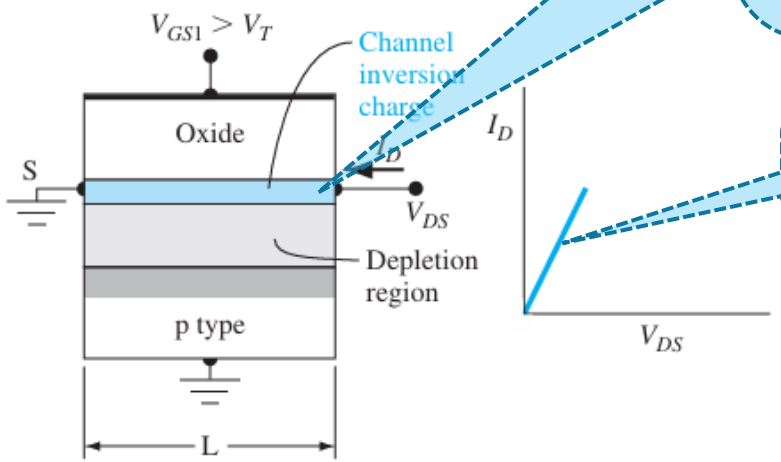
(1)$V_{DS} > 0$(V_{DS} is very small)



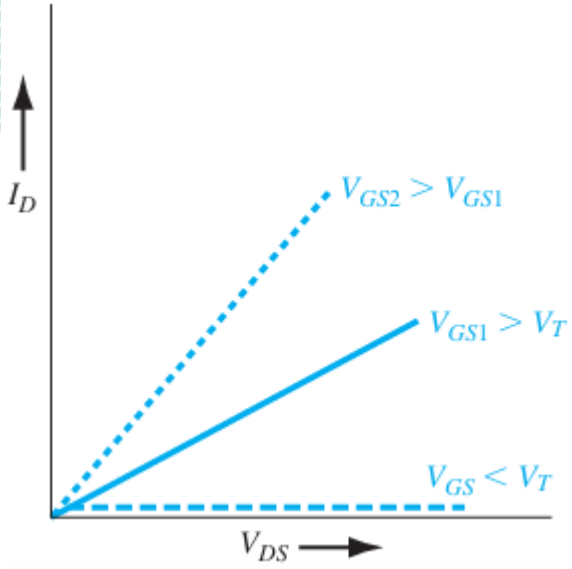
The thickness of the inversion channel layer in the figure qualitatively indicates the relative charge density, which is essentially constant along the entire channel length for this case.

$$I_D = g_d V_{DS}$$

$$g_d = \frac{W}{L}\mu_n |Q'_n|$$

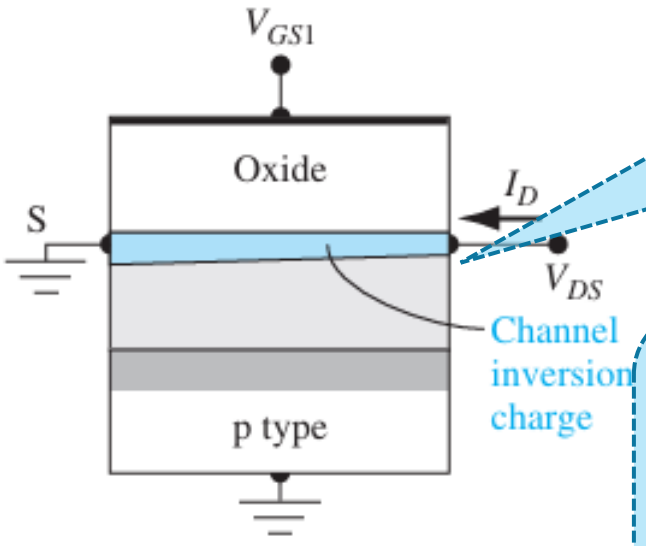


Linear region



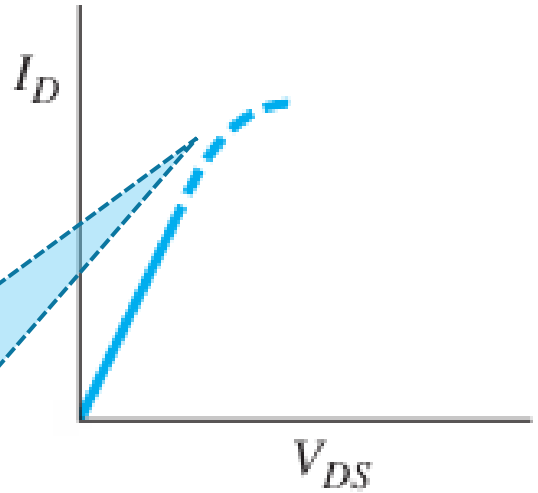
10.3.2 Current–Voltage Relationship—Concepts

(2)$0 < V_{DS} < V_{GS} - V_T$ ⇨ $V_{GD} = V_{GS} - V_{DS} > V_T$



The induced inversion charge density near the drain decreases.

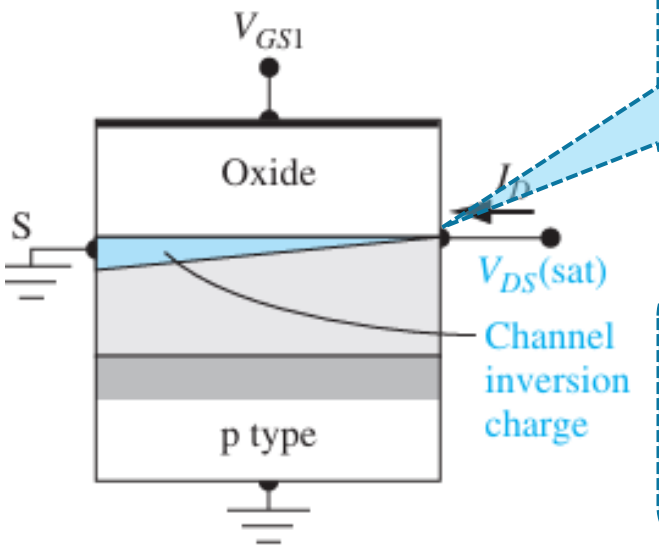
The incremental conductance of the channel at the drain decreases, which then means that the slope of the I_D versus V_{DS}curve will decrease.



10.3.2 Current–Voltage Relationship—Concepts

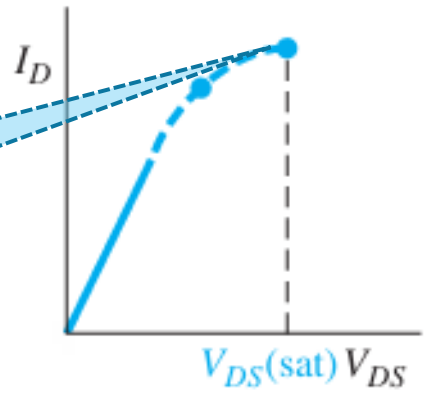
(3)$V_{DS} = V_{GS} - V_T$ ⇨ $V_{GD} = V_T$

VDS (sat) is the drain-to-source voltage producing zero inversion charge density at the drain terminal.



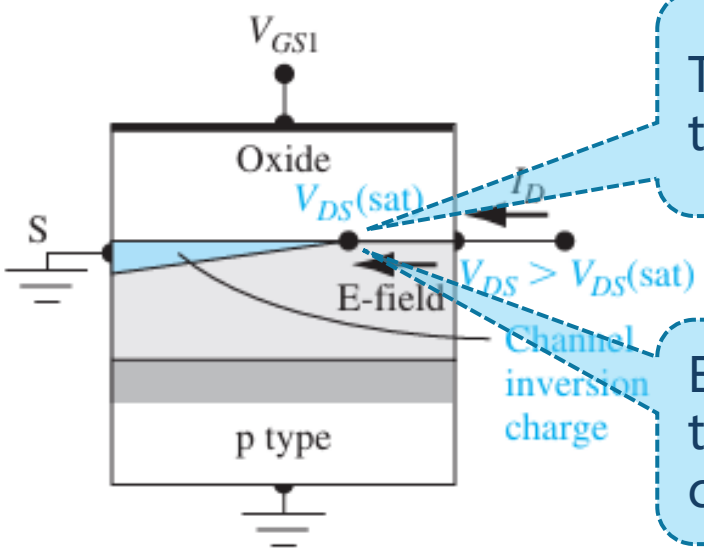
The inversion charge density is zero at the drain terminal. (pinch off)

The incremental conductance at the drain is zero, which means that that the slope of the I_D versus V_{DS}curve is zero.



10.3.2 Current–Voltage Relationship—Concepts

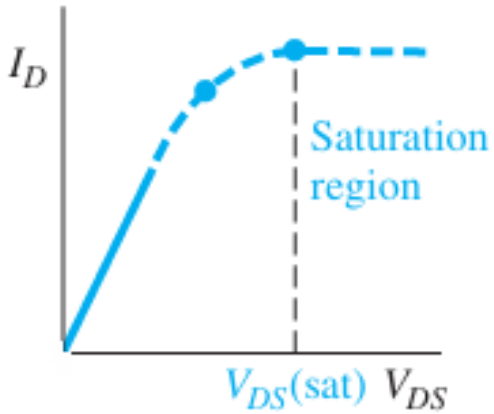
(4)$V_{DS} > V_{GS} - V_T$ ⇨ $V_{GD} < V_T$



The pinch-off point moves toward the source terminal

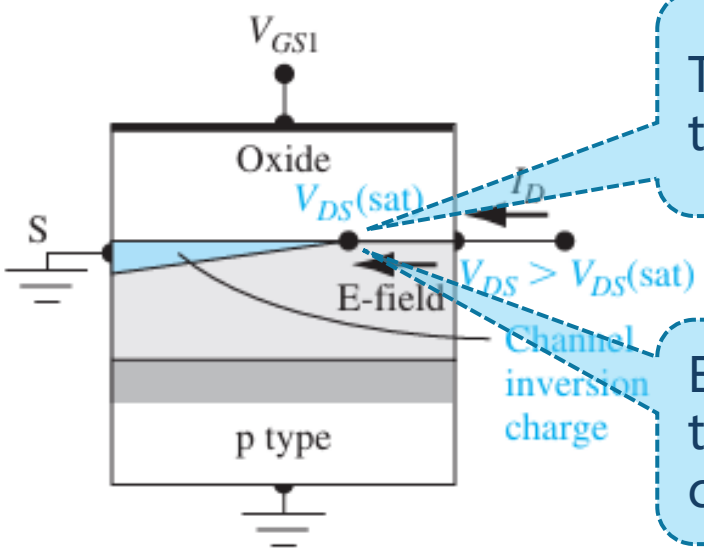
Electrons enter the channel at the source, travel through the channel toward the drain.

The voltage between the pinch-off point and the source terminal maintains $V_{DS}(sat)$.



10.3.2 Current–Voltage Relationship—Concepts

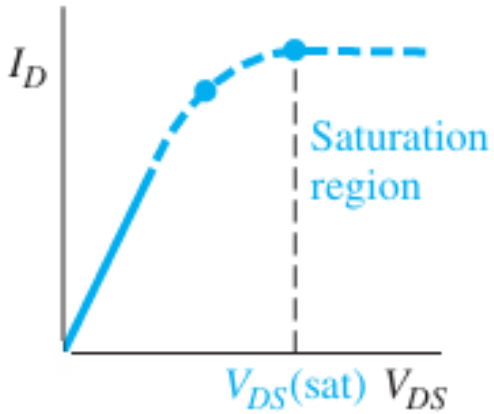
(4)$V_{DS} > V_{GS} - V_T$ ⇨ $V_{GD} < V_T$



The pinch-off point moves toward the source terminal

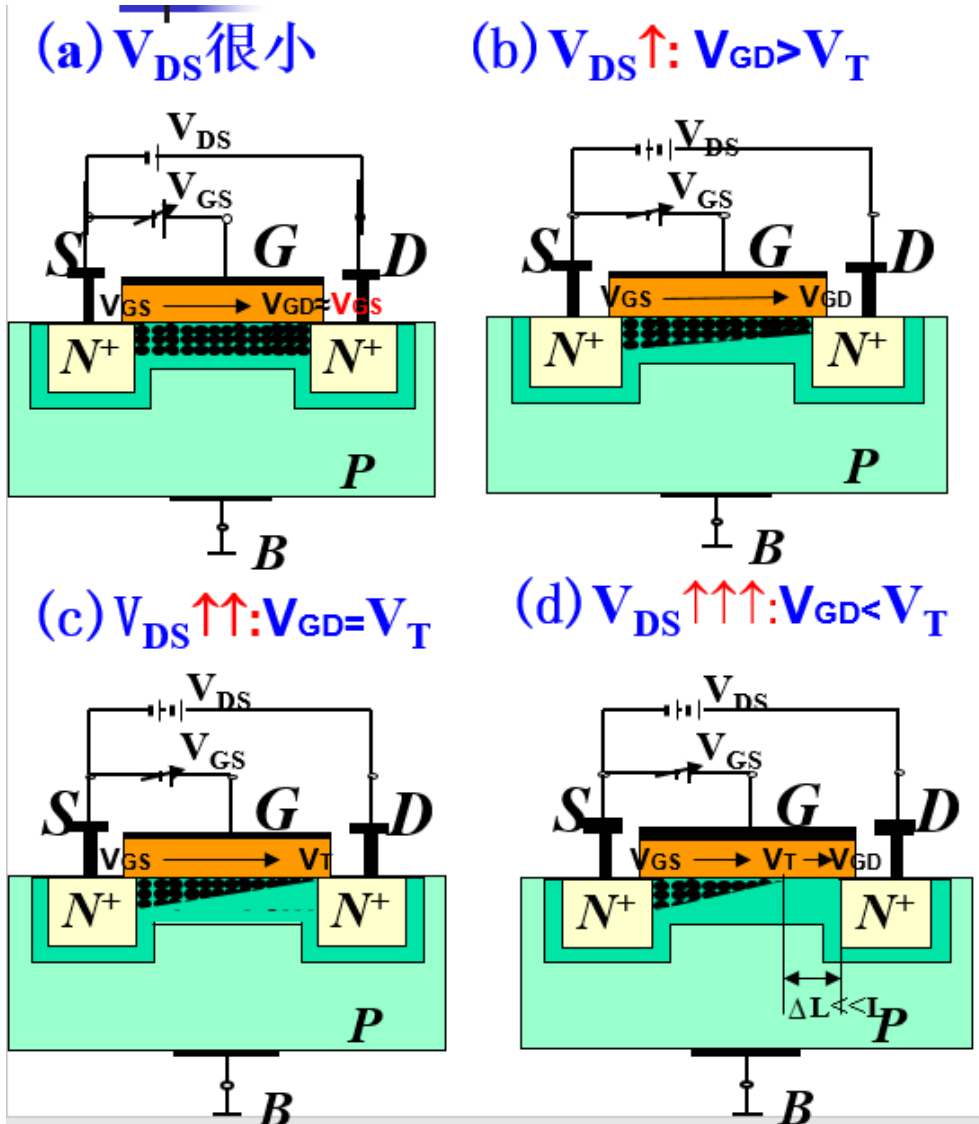
Electrons enter the channel at the source, travel through the channel toward the drain.

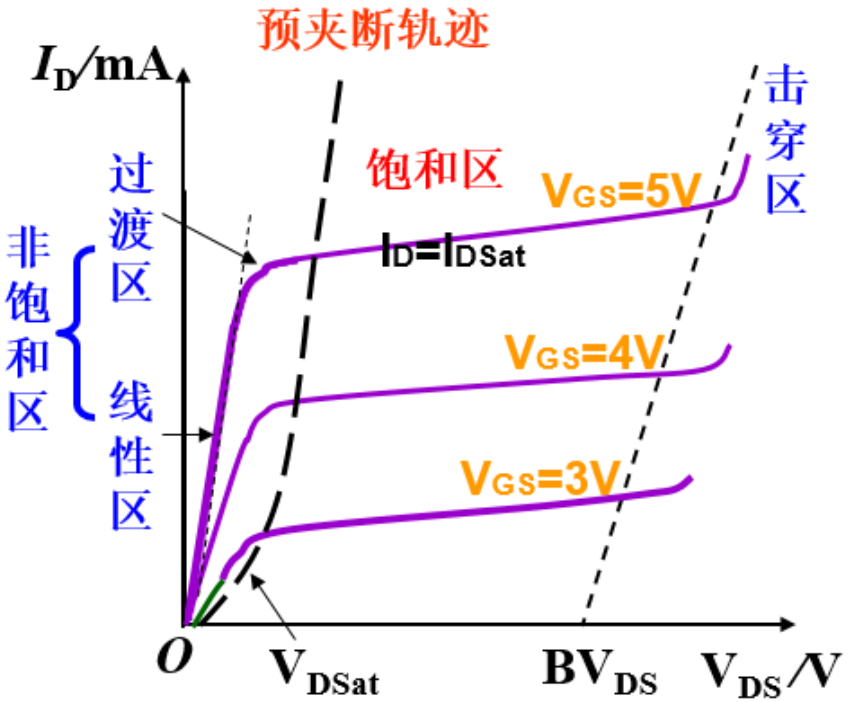
The voltage between the pinch-off point and the source terminal maintains $V_{DS}(sat)$.



10.3.2 Current–Voltage Relationship—Concepts

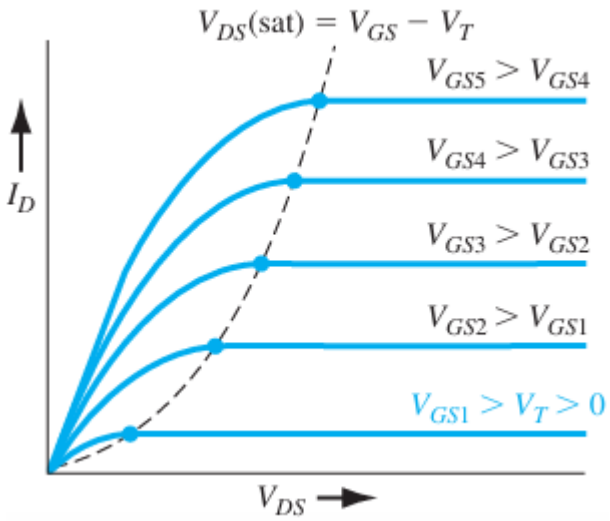
3. Output characteristic



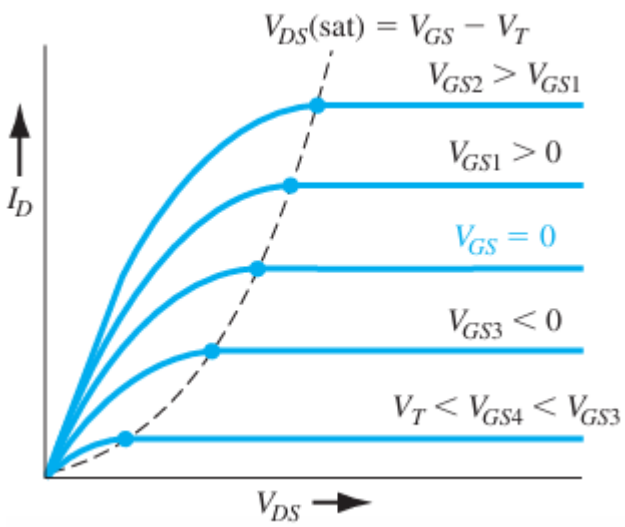


10.3.2 Current–Voltage Relationship—Concepts

3. Output characteristic



Enhancement mode



Depletion mode

Nonsaturation region:

$$I_D = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$$

$$I_D = \frac{k_n'}{2} \cdot \frac{W}{L} \cdot \left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$$

$$I_D = K_n\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$$

saturation region:

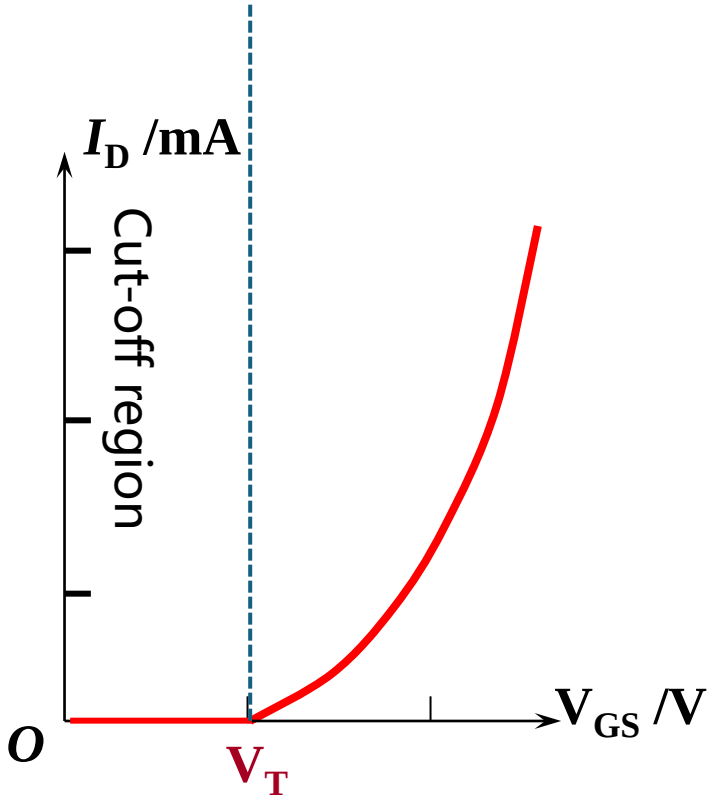
$$I_D = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$

$$I_D = \frac{k_n'}{2} \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2$$

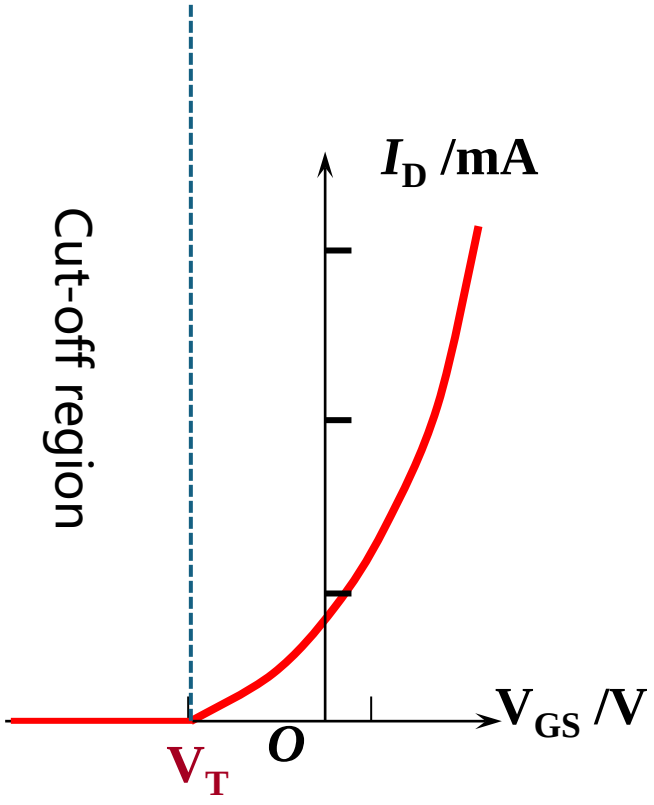
$$I_D = K_n(V_{GS} - V_T)^2$$

10.3.2 Current–Voltage Relationship—Concepts

4. Transfer characteristic



Enhancement mode

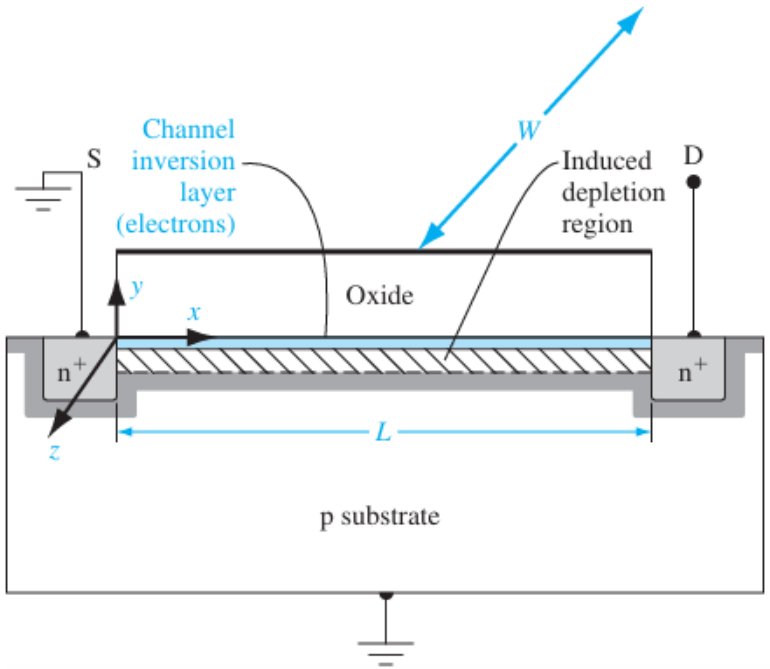


Depletion mode

*10.3.3 Current–Voltage Relationship—Mathematical Derivation

Assumptions:

1. The current in the channel is due to drift rather than diffusion.
2. There is no current through the gate oxide.
3. A gradual channel approximation is used in which $\partial E_y \ll \partial E_x$. This approximation means that E_x is essentially a constant.
4. Any fixed oxide charge is an equivalent charge density at the oxide semiconductor interface.
5. The carrier mobility in the channel is constant.



*10.3.3 Current–Voltage Relationship—Mathematical Derivation

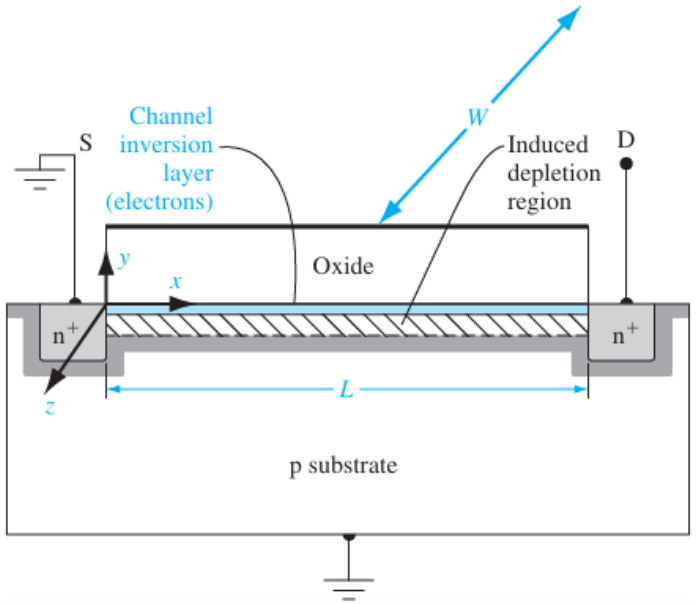
$$J_x = \sigma E_x$$

Ohm' s law
$\sigma = e\mu_n n(y)$

$$I_x = \int_y \int_z J_x dydz = \int_0^W \mu_n E_x \int_y en(y)dydz = -W\mu_n Q'_n E_x = W\mu_n Q'_n \frac{dV_x}{dx}$$

$$Q'_n = -\int en(y)dy$$

$$E_x = -\frac{dV_x}{dx}$$



*10.3.3 Current–Voltage Relationship—Mathematical Derivation

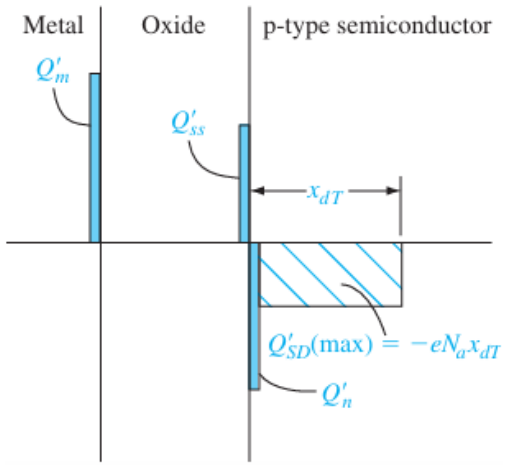
Charge neutrality: $Q'_m + Q'_{ss} + Q'_n + Q'_{SD}(max) = 0$

$$Q_T = [Q'_{ss} + Q'_n + Q'_{SD}(max)]Wdx$$

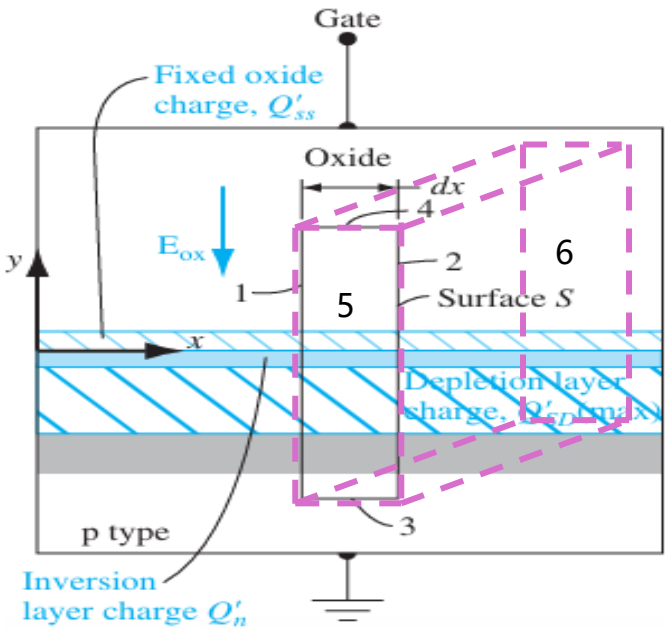
Gauss' s law: $\oint_s \epsilon E_n dS = Q_T$

$$\oint_s \epsilon E_n dS = \oint_{s1} \epsilon_1 E_1 dS + \oint_{s2} \epsilon_2 E_2 dS + \oint_{s3} \epsilon_3 E_3 dS + \oint_{s4} \epsilon_4 E_4 dS$$

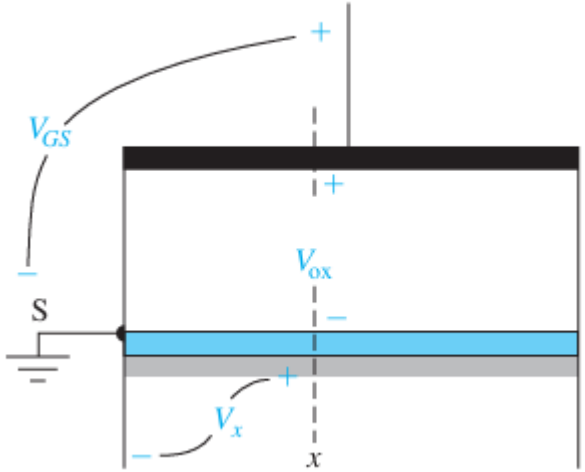
$$+ \oint_{s5} \epsilon_5 E_5 dS + \oint_{s6} \epsilon_6 E_6 dS = \oint_{s4} \epsilon_4 E_4 dS = -\epsilon_{ox} E_{ox} W dx = Q_T$$

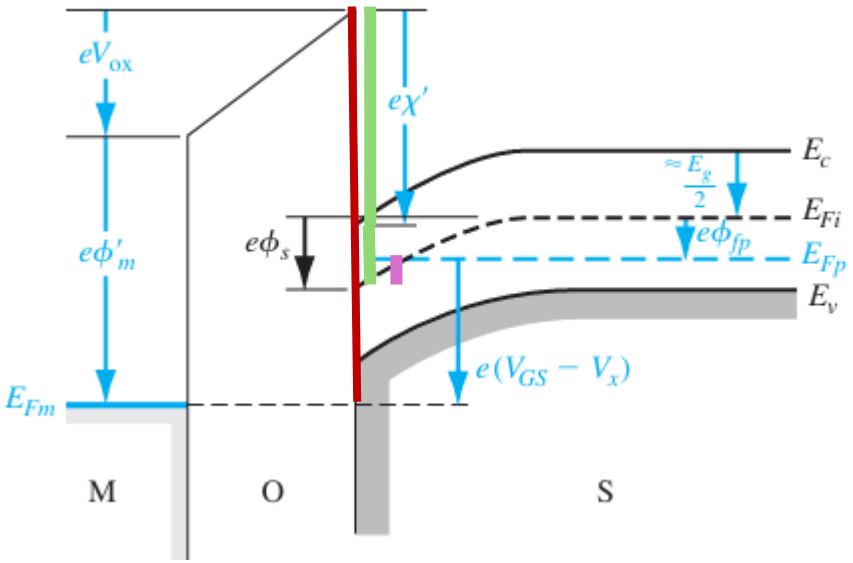


$$-\epsilon_{ox} E_{ox} = Q'_{ss} + Q'_n + Q'_{SD}(max)$$



*10.3.3 Current–Voltage Relationship—Mathematical Derivation





$$\phi_s = 2\phi_{fp}$$

$$E_{fp} - E_{fm} = e(\phi'_m + V_{ox}) - e\left(\chi' + \frac{E_g}{2e} - \phi_s + \phi_{fp}\right)$$

$$\phi_{ms} = \phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)$$

$$E_{fp} - E_{fm} = e(V_{GS} - V_x)$$

$$\Rightarrow \quad V_{GS} - V_x = V_{ox} + \phi_{ms} + 2\phi_{fp}$$

*10.3.3 Current–Voltage Relationship—Mathematical Derivation

$$V_{GS} - V_x = V_{ox} + \phi_{ms} + 2\phi_{fp}$$

$$E_{ox} = \frac{V_{ox}}{t_{ox}}$$

$$V_{ox} = (V_{GS} - V_x) - (\phi_{ms} + 2\phi_{fp})$$

$$E_{ox} = \frac{(V_{GS} - V_x) - (\phi_{ms} + 2\phi_{fp})}{t_{ox}}$$

$$-\epsilon_{ox} E_{ox} = Q'_{ss} + Q'_n + Q'_{SD}(max)$$

$$-\epsilon_{ox} E_{ox} = \frac{-\epsilon_{ox}}{t_{ox}}\left[(V_{GS} - V_x) - (\phi_{ms} + 2\phi_{fp})\right] = Q'_{ss} - |Q'_n| - |Q'_{SD}(max)|$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$|Q'_n| = -|Q'_{SD(\max)}| + Q_{ss}' + \frac{\varepsilon_{ox}}{t_{ox}}\left[(V_{GS} - V_x) - (2\phi_{fp} + \phi_{ms})\right]$$

Threshold $voltage(V_T)$

$$|Q'_n| = \frac{\epsilon_{ox}}{t_{ox}}\left\{V_{GS} - V_x - \left[(|Q'_{SD}(max)| - Q'_{ss})\frac{t_{ox}}{\epsilon_{ox}} + 2\phi_{fp} + \phi_{ms}\right]\right\}$$

$$Q'_n = -C_{ox}(V_{GS} - V_x - V_T)$$

$$I_x = -W\mu_n Q'_n E_x$$

$$I_x = -W\mu_n C_{ox}(V_{GS} - V_x - V_T)\frac{dV_x}{dx}$$

*10.3.3 Current–Voltage Relationship—Mathematical Derivation

$$I_x = -W\mu_n C_{ox}(V_{GS} - V_x - V_T)\frac{dV_x}{dx}$$

$$\int_0^L I_x dx = -W\mu_n C_{ox}\int_{V_{ox}(0)}^{V_{ox}(L)} [(V_{GS} - V_T) - V_x]\, dV_x$$

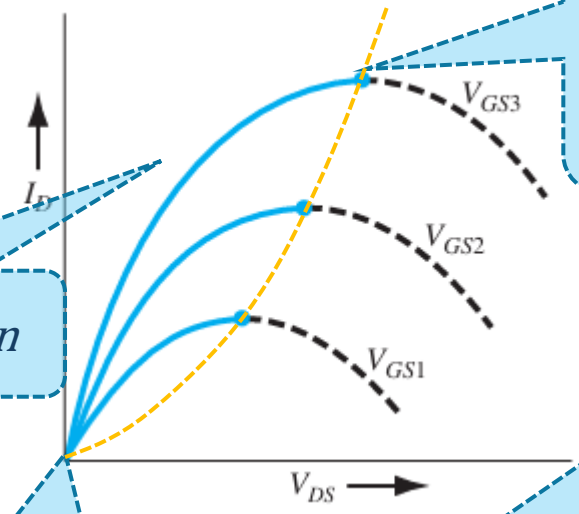
$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

$$I_D = k'_n\frac{W}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

$$I_D = k'_n\frac{W}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

$V_{GS} \geq V_T$

$0 \leq V_{DS} \leq V_{GS} - V_T$



$$\frac{\partial I_D}{\partial V_{DS}} = 0$$

$$V_{DS} = V_{GS} - V_T = V_{DS}(sat)$$

nonsaturation region

In the linear region:

$$I_D = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)V_{DS}$$

$V_{DS} \ll V_{GS} - V_T$

in the saturation region:

$$I_D(sat) = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS}(sat) - V_{DS}^2(sat)]$$

$$I_D(sat) = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$

$V_{DS} \geq V_{GS} - V_T$

*10.3.3 Current–Voltage Relationship—Mathematical Derivation

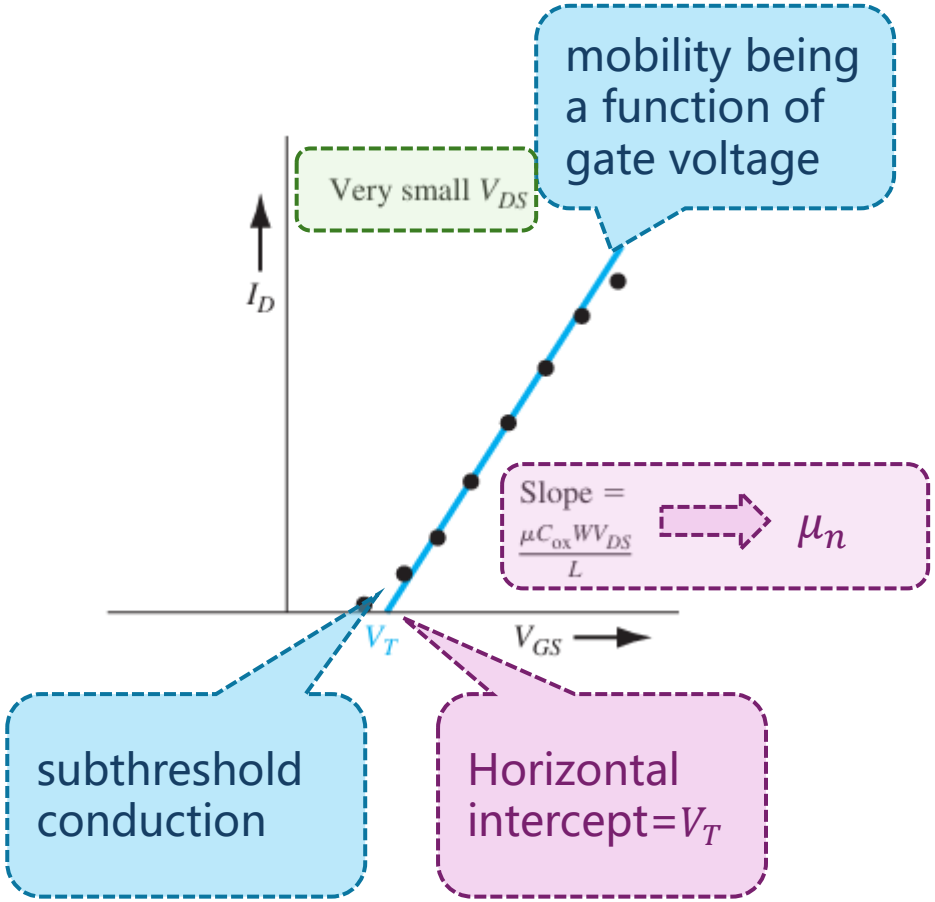
We can use the I–V relations to experimentally determine the mobility and threshold voltage parameters.

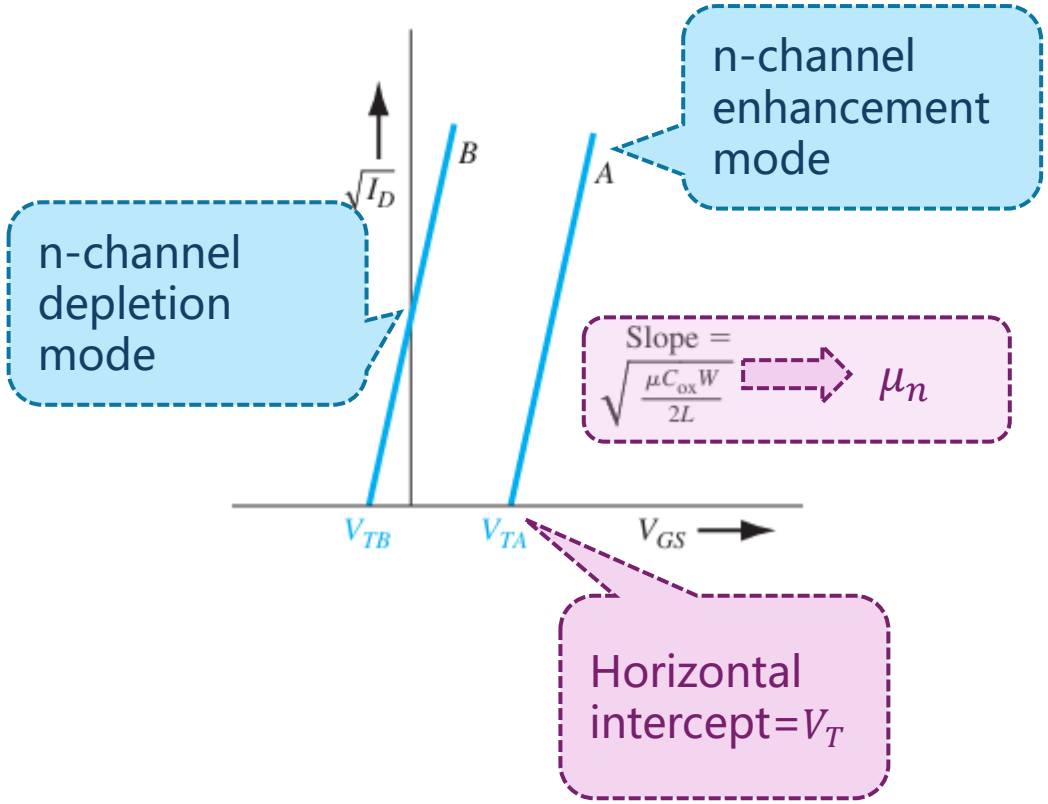
In the linear region:

$$I_D = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)V_{DS}$$

in the saturation region:

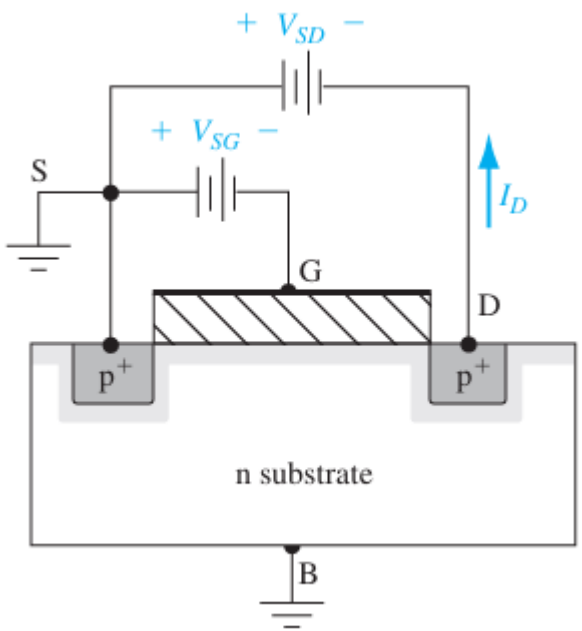
$$\sqrt{I_D(sat)} = \sqrt{\frac{W\mu_n C_{ox}}{2L}}(V_{GS} - V_T)$$





*10.3.3 Current–Voltage Relationship—Mathematical Derivation

p-channel enhancement mode MOSFET



In the linear region:

$$I_D = \frac{W\mu_n C_{ox}}{L}(V_{SG} + V_T)V_{DS}$$

in the non-saturation region:

$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{SG} + V_T)V_{DS} - V_{DS}^2]$$

in the saturation region:

$$I_D(sat) = \frac{W\mu_n C_{ox}}{2L}(V_{SG} + V_T)^2$$

$$V_{SD}(sat) = V_{SG} + V_T$$

10.3.4 Transconductance (transistor gain)

The MOSFET transconductance is defined as the change in drain current with respect to the corresponding change in gate voltage.

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}=constant}$$

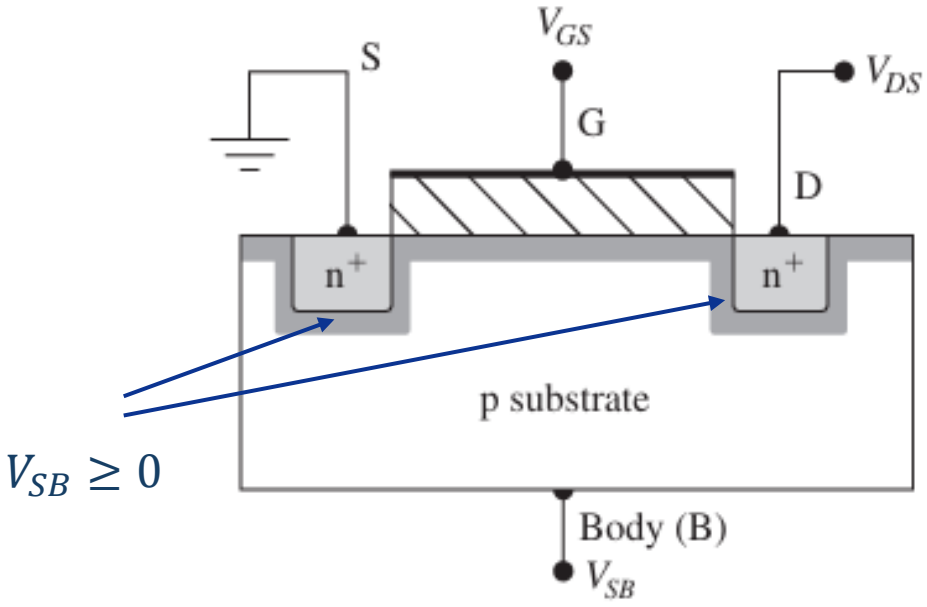
- *in the non-saturation region:* $g_{mL} = \frac{W\mu_n C_{ox}}{L} V_{DS}$
  - $\propto V_{DS}$
  - g_{mL} *is independent of* V_{GS}
- *in the saturation region:* $g_{ms} = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)$
  - $g_{ms} \propto (V_{GS} - V_T)$
  - g_{ms} *is independent of* V_{DS}

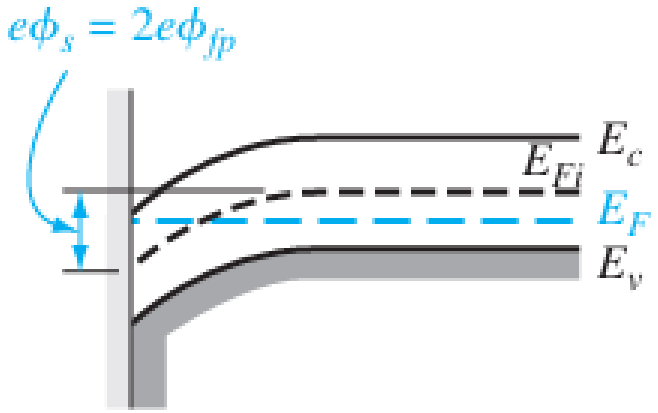
The g_m *is function of*

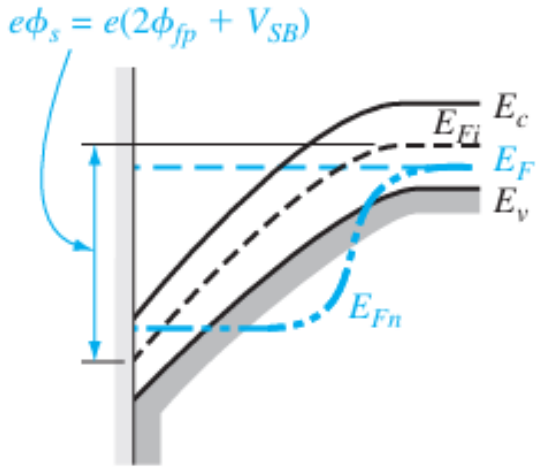
- the geometry of the device
  - $W \Uparrow$
  - $L \Downarrow$
  - $t_{ox} \Downarrow$

  $\Rightarrow g_m \Uparrow$
- carrier mobility(μ_n)
- threshold voltage

10.3.5 Substrate Bias Effects







$V_{SB} = 0$

$E_{Fn} = E_F$

$\phi_S = 2\phi_{fp}$

$\phi'_{SD}(max) = -eN_a x_d$

$= -\sqrt{2e\epsilon_s N_a (2\phi_{fp})}$

$V_{SB} > 0$

$E_{Fn} = E_F - eV_{SB}$

$\phi_S = 2\phi_{fp} + V_{SB}$

$\phi'_{SD} = -eN_a x_d$

$= -\sqrt{2e\epsilon_s N_a (2\phi_{fp} + V_{SB})}$

$$\Delta\phi'_{SD} = -\sqrt{2e\epsilon_s N_a}\left[\sqrt{(2\phi_{fp} + V_{SB})} - \sqrt{2\phi_{fp}}\right]$$

10.3.5 Substrate Bias Effects

$$\Delta\phi'_{SD} = -\sqrt{2e\epsilon_s N_a}\left[\sqrt{(2\phi_{fp} + V_{SB})} - \sqrt{2\phi_{fp}}\right]$$

$$V_T = \frac{|Q'_{SD}(max)|}{C_{ox}} - (-Q'_{SS})\frac{t_{ox}}{\epsilon_{ox}} + 2\phi_{fp} + \phi_{ms}$$

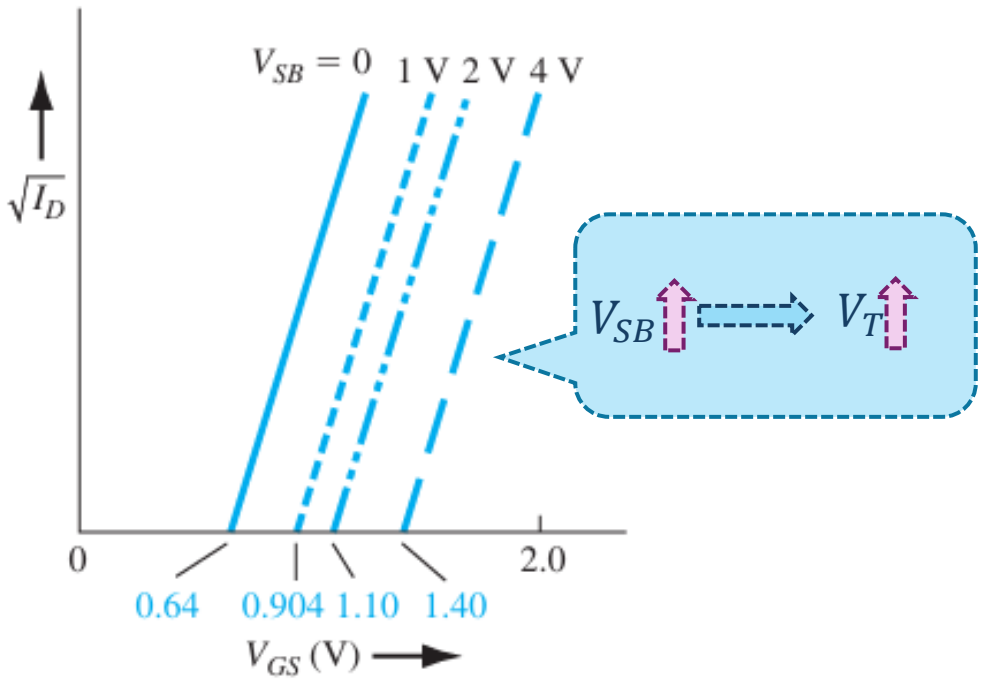
$$\Delta V_T = V_T(V_{SB} > 0) - V_T(V_{SB} = 0) > 0$$

$$\Delta V_T = -\frac{\Delta\phi'_{SD}}{C_{ox}} = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}\left[\sqrt{(2\phi_{fp} + V_{SB})} - \sqrt{2\phi_{fp}}\right]$$

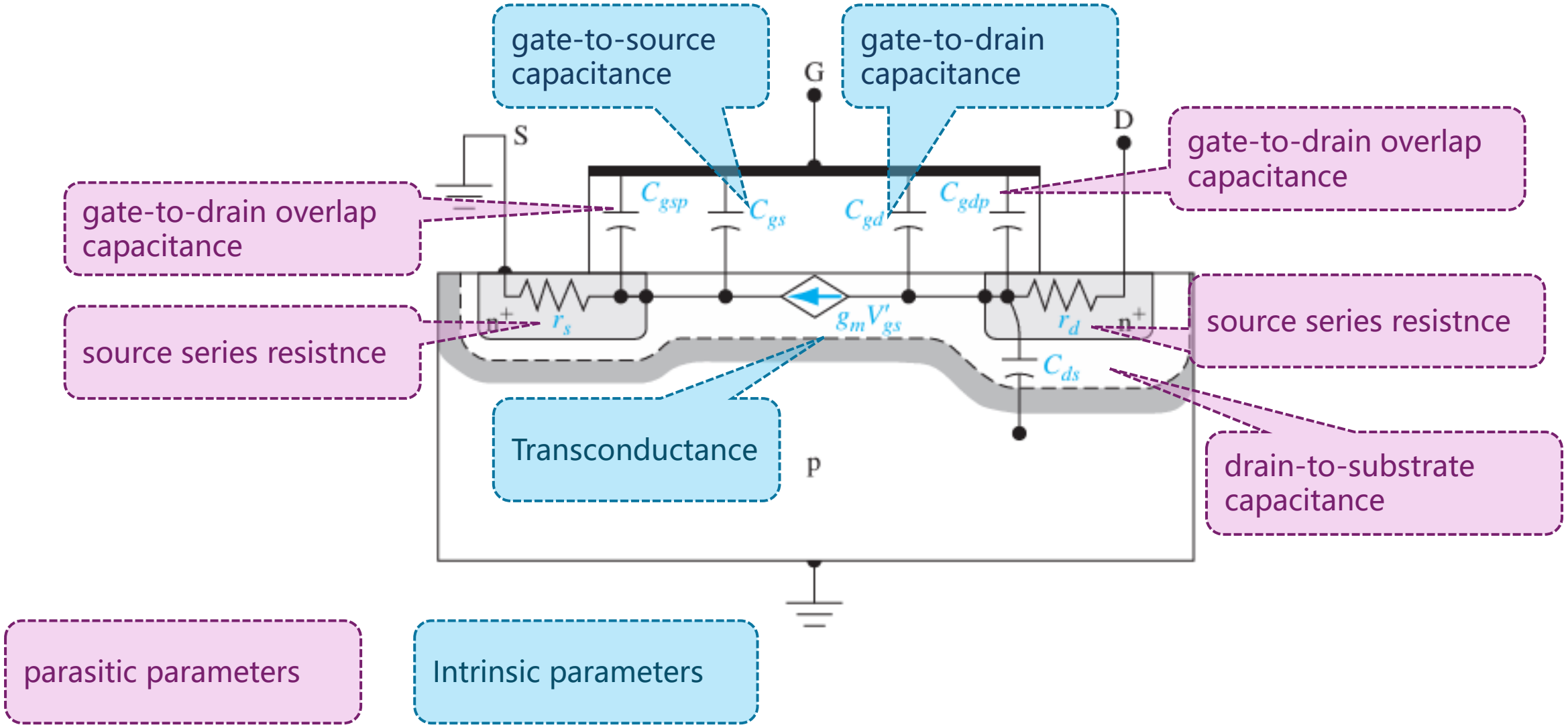
$$\gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$$

body-effect coefficient

$$\Delta V_T = \gamma\left[\sqrt{(2\phi_{fp} + V_{SB})} - \sqrt{2\phi_{fp}}\right]$$

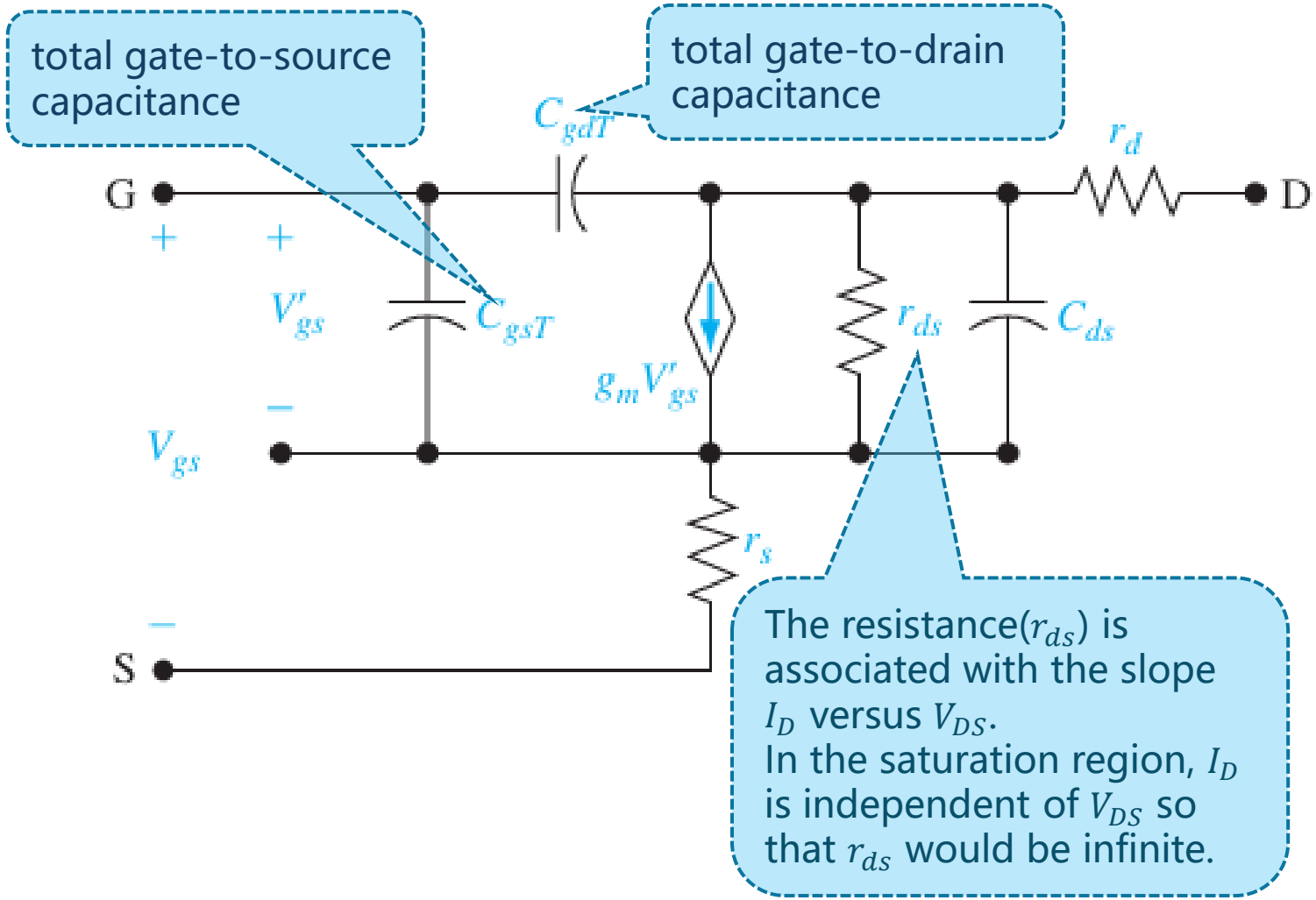


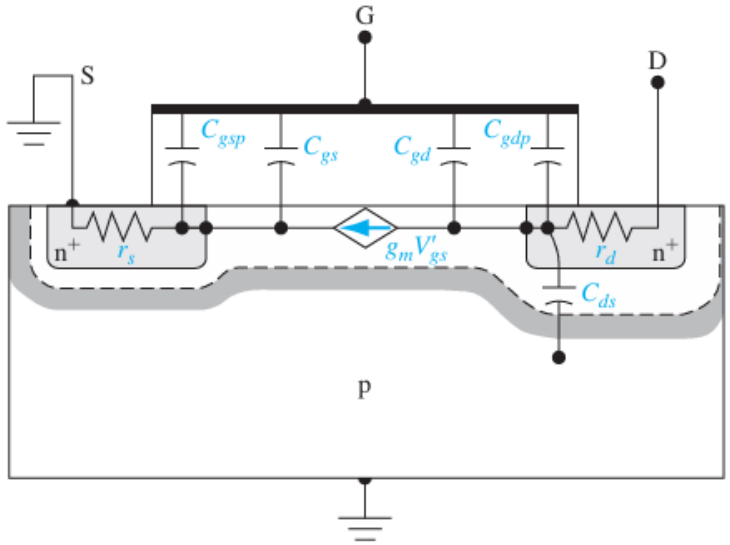
10.4 | FREQUENCY LIMITATIONS
10.4.1 Small-Signal Equivalent Circuit



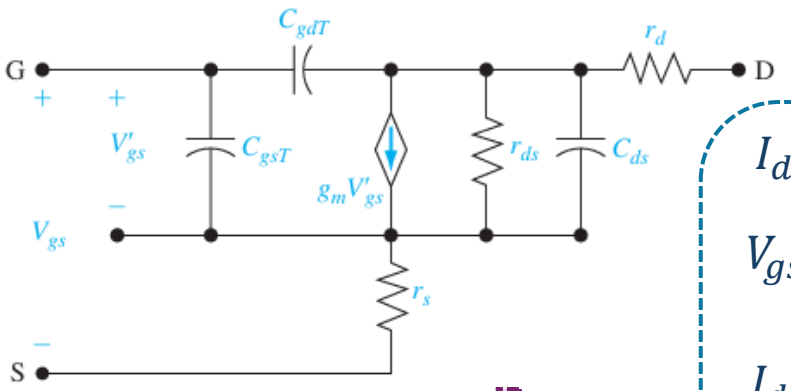
10.4.1 Small-Signal Equivalent Circuit

Small-signal equivalent circuit of a common source n-channel MOSFET





10.4.1 Small-Signal Equivalent Circuit

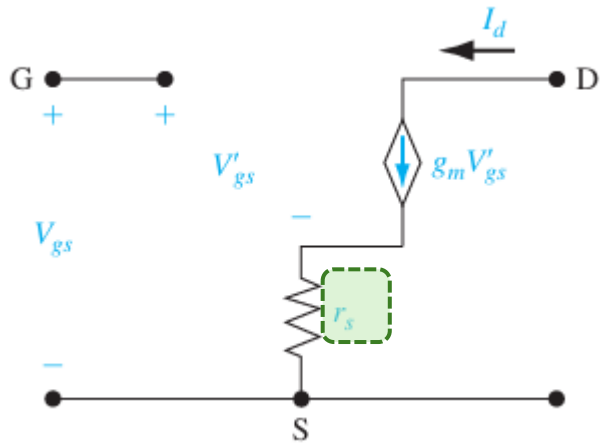


$$I_d = g_m V'_{gs}$$

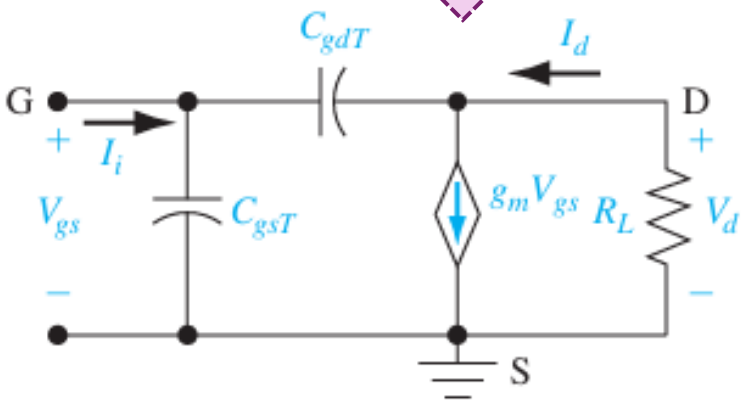
$$V_{gs} = V'_{gs} + (g_m V'_{gs}) r_s = (1 + g_m r_s) V'_{gs} > V'_{gs}$$

$$I_d = \left(\frac{g_m}{1+g_m r_s}\right) V_{gs} = g'_m V_{gs}$$

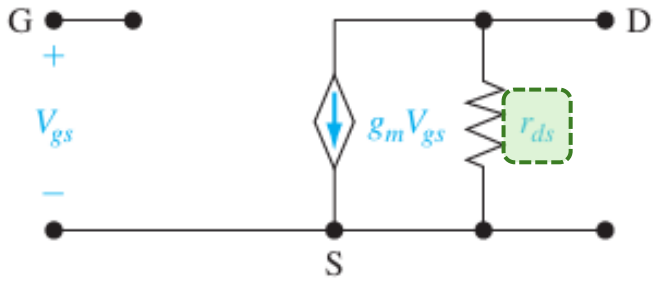
$$g'_m = \frac{g_m}{1+g_m r_s} < g_m$$



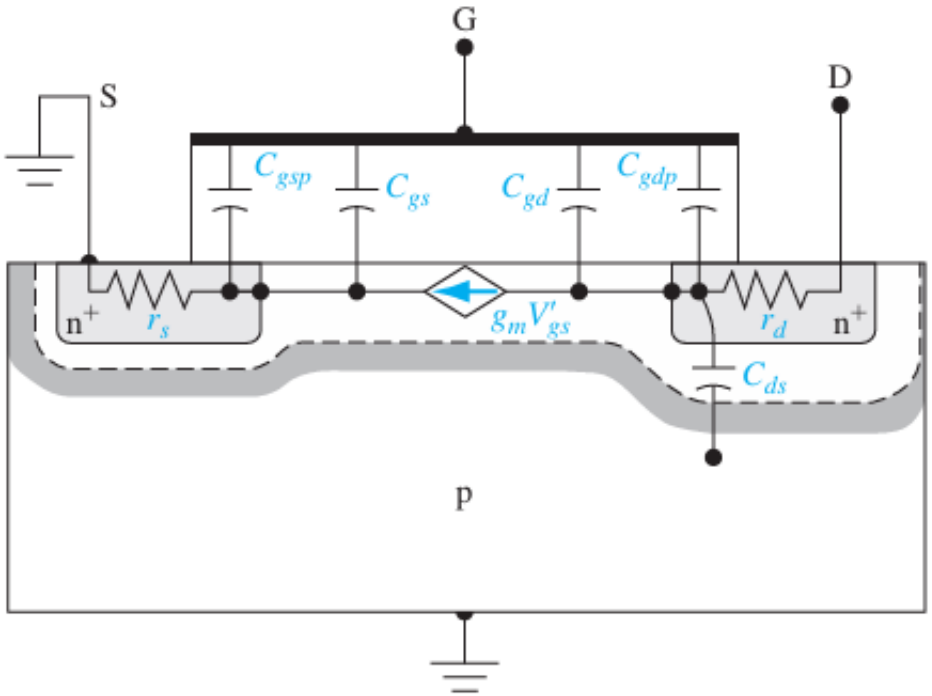
Intrinsic parameters



Low-frequency



10.4.2 Frequency Limitation Factors and Cutoff Frequency



1.The first factor is the channel transit time.

$$\tau_t = \frac{L}{v_{sat}}$$

Cutoff frequency

$$f_t = \frac{1}{\tau_t}$$

τ_t is usually not the limiting factor in the frequency responses of MOSFETs.

$v_{sat} = 10^7 cm/s$

$L = 1\mu m$

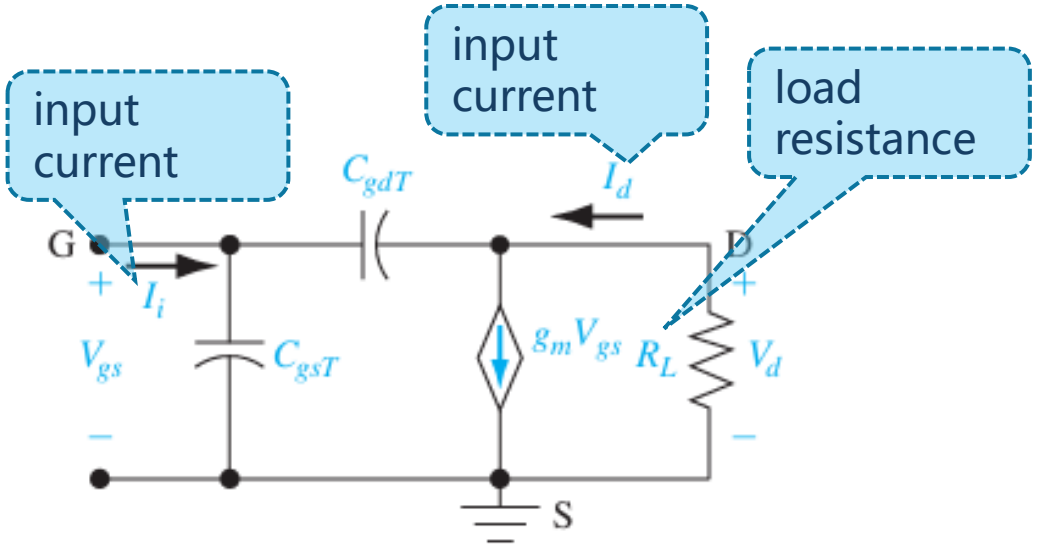
$\tau_t = 1ps$

$f_t = 100\text{GHz}$

2.The second limiting factor is the gate or capacitance charging time.

10.4.2 Frequency Limitation Factors and Cutoff Frequency

2.The second limiting factor is the gate or capacitance charging time.



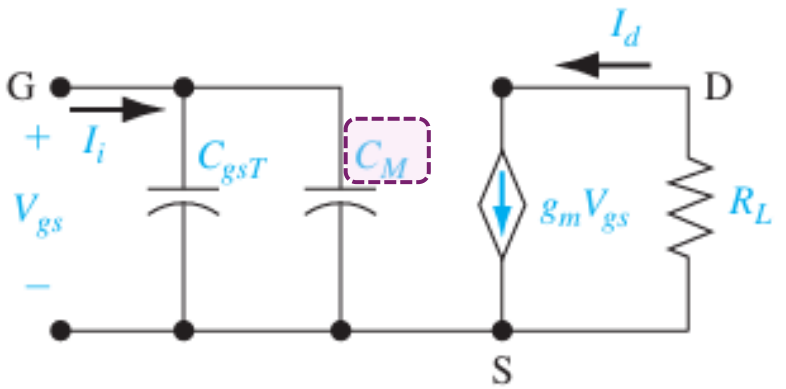
$$\begin{cases} I_i = j\omega C_{gsT} V_{gs} + j\omega C_{gdT}(V_{gs} - V_d) \\ \dfrac{V_d}{R_L} + g_m V_{gs} + j\omega C_{gdT}(V_d - V_{gs}) = 0 \end{cases}$$

$\omega R_L C_{gdT} << 1$

$$I_i = j\omega\left[C_{gsT} + C_{gdT}\left(\frac{1 + g_m R_L}{1 + j\omega R_L C_{gdT}}\right)\right]V_{gs}$$

$$I_i = j\omega\left[C_{gsT} + C_{gdT}(1 + g_m R_L)\right]V_{gs}$$

Miller capacitance
$C_M = C_{gdT}(1 + g_m R_L)$



$$= j\omega(C_{gsT} + C_M)V_{gs}$$

drain overlap capacitance

In the saturation region, $C_{gd} = 0$, and the parasitic capacitance is multiplied by the gain of the transistor

10.4.2 Frequency Limitation Factors and Cutoff Frequency

$$\begin{cases} I_i = j\omega(C_{gsT} + C_M)V_{gs} \\ I_d = g_m V_{gs} \end{cases} \Rightarrow \left|\frac{I_i}{I_d}\right| = \frac{g_m}{2\pi f(C_{gsT} + C_M)}$$

current gain

$$f_T = f_{\left|\frac{I_i}{I_d}\right|=1} \Rightarrow \left|\frac{I_i}{I_d}\right| = \frac{g_m}{2\pi f_T(C_{gsT} + C_M)} = 1 \Rightarrow f_T = \frac{g_m}{2\pi(C_{gsT} + C_M)} = \frac{g_m}{2\pi C_G}$$

Transconductance

equivalent input gate capacitance

In the ideal MOSFET:

saturation region

$$\left.\begin{cases} C_{gsp} = C_{gdp} = 0 \\ C_{gd} = 0 \end{cases}\right\} C_M = C_{gdT}(1 + g_m R_L) = 0 \Rightarrow C_G = C_{gsT} = C_{gs} = C_{ox}WL$$

$$g_{ms} = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)$$

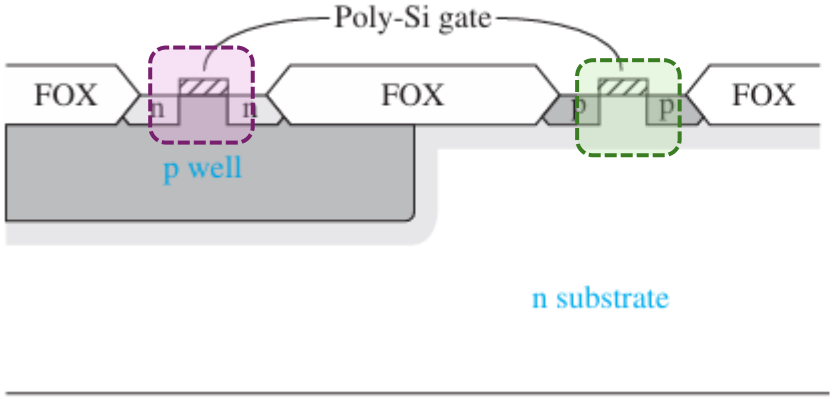
$$\Rightarrow f_T = \frac{\frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)}{2\pi(C_{ox}WL)}$$

$$f_T \uparrow = \frac{\mu_n(V_{GS} - V_T)}{2\pi L^2}$$

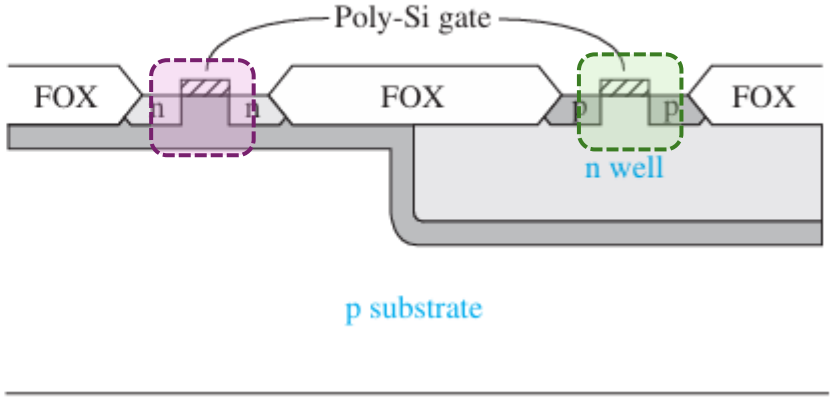
$\mu_n \uparrow$ $C_{gsp} \downarrow$ $L \downarrow$ $g_{ms} \uparrow$

$C_{gdp} \downarrow$

*10.5 | THE CMOS TECHNOLOGY



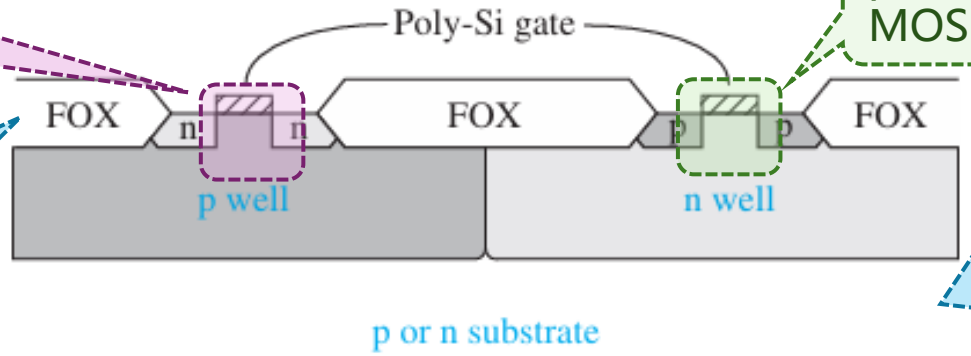
p well



n well

n channel MOSFET

p channel MOSFET



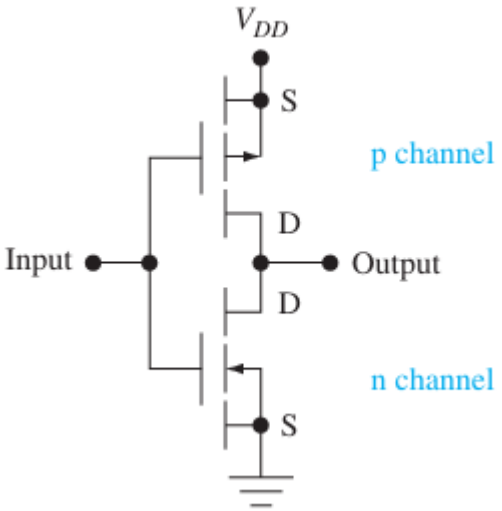
Field oxide is a relatively thick oxide separating the devices.

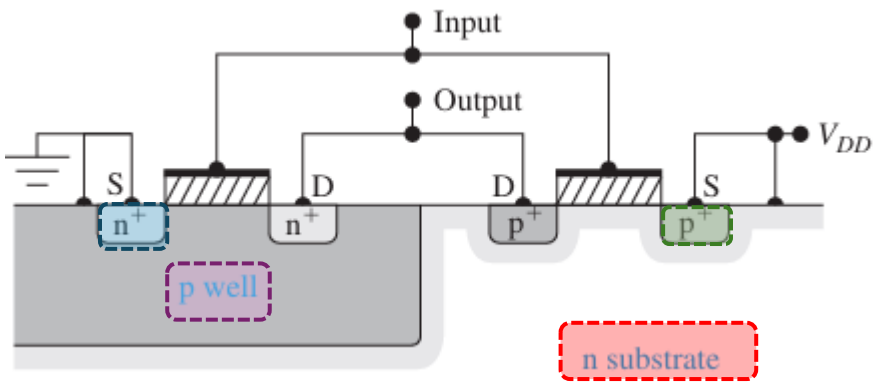
The twin-well CMOS process, allows both the p-well and n-well regions to be optimally doped to control the threshold voltage and trans conductance of each transistor.

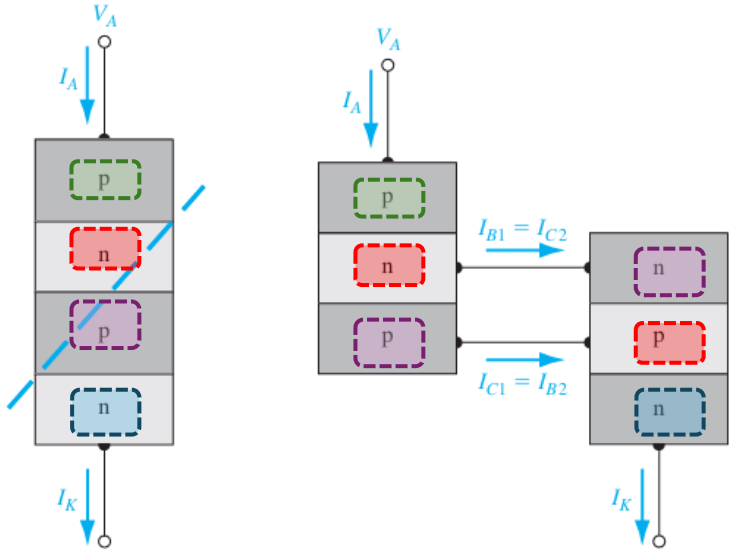
twin well

*10.5 | THE CMOS TECHNOLOGY

Latch-up refers to a high-current, low-voltage condition that may occur in a four-layer pnpn structure.







Latch-up can be prevented if the product n p is less than unity.

- "kill" the minority carrier lifetime by gold doping or neutron irradiation
- proper circuit layout techniques.
- silicon-on-insulator technology

Exercises

10.2，10.7 、10.10 、10.23、10.35，10.46，10.48，10.51